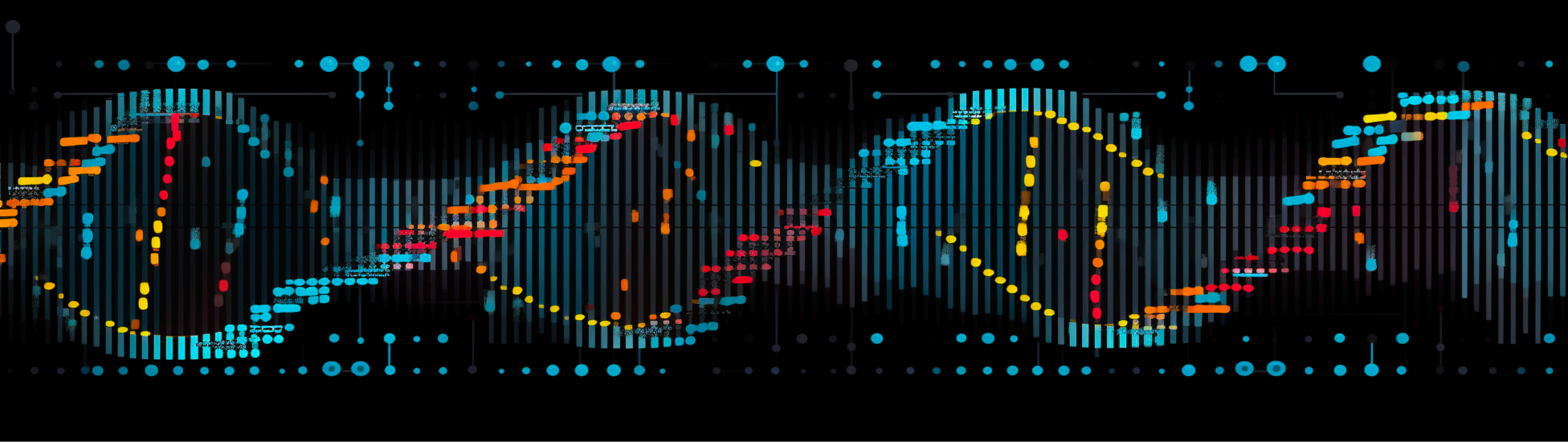


Introduction to bioinformatics

Chapter3 part2 Data and Resource

Jiaxing Chen

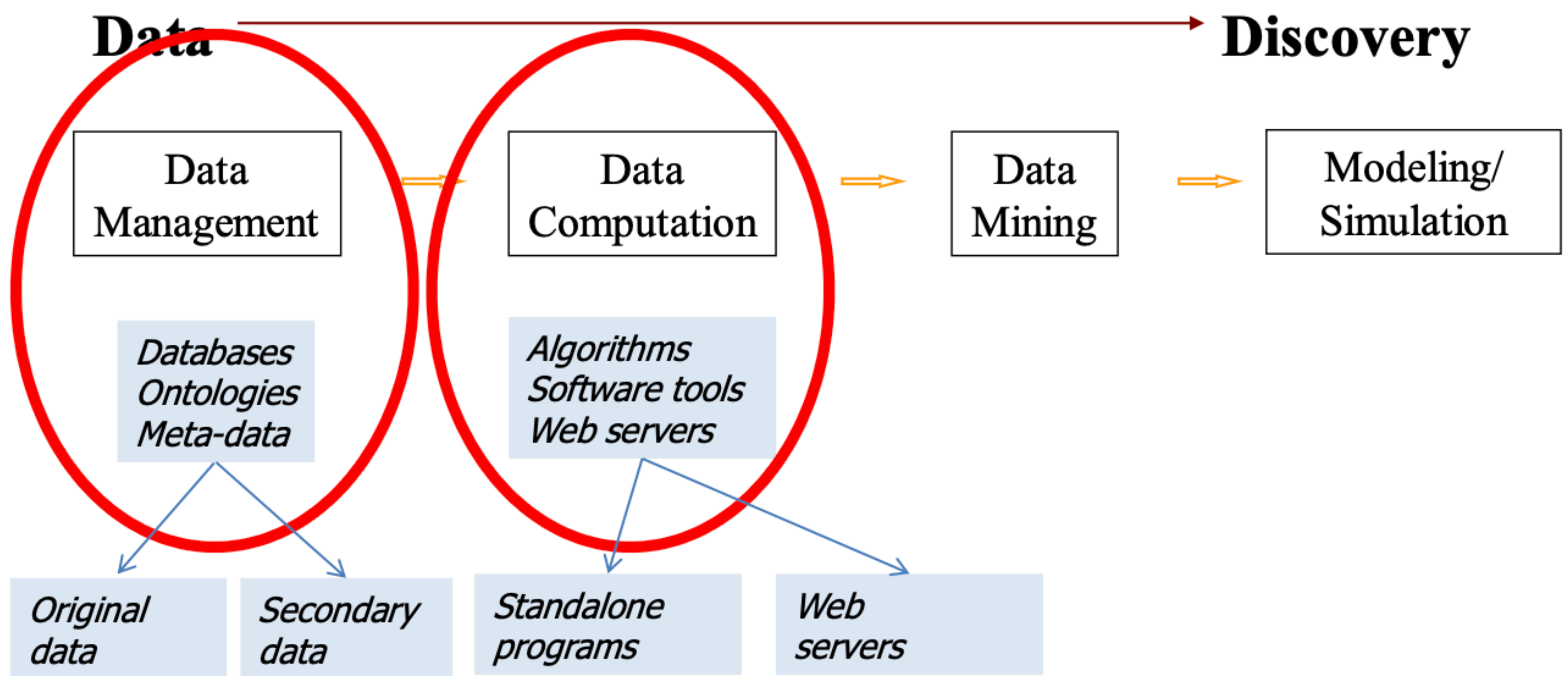
Outline

- **Biological file format**
- Resource
  - NCBI
  - EBI
  - UCSC
  - PDB
  - …

Definition of a biological database/resources

- The NAR Database Issue collects publications of established databases in the field
- Collection of data (and metadata) in the related format
    - structured
    - searchable (indexed)
    - updated periodically
    - entries mapped to unique identifiers, and cross-referenced
- Includes associated software necessary for DB access, update, search, visualisation (web)

The –informatics in Bioinformatics



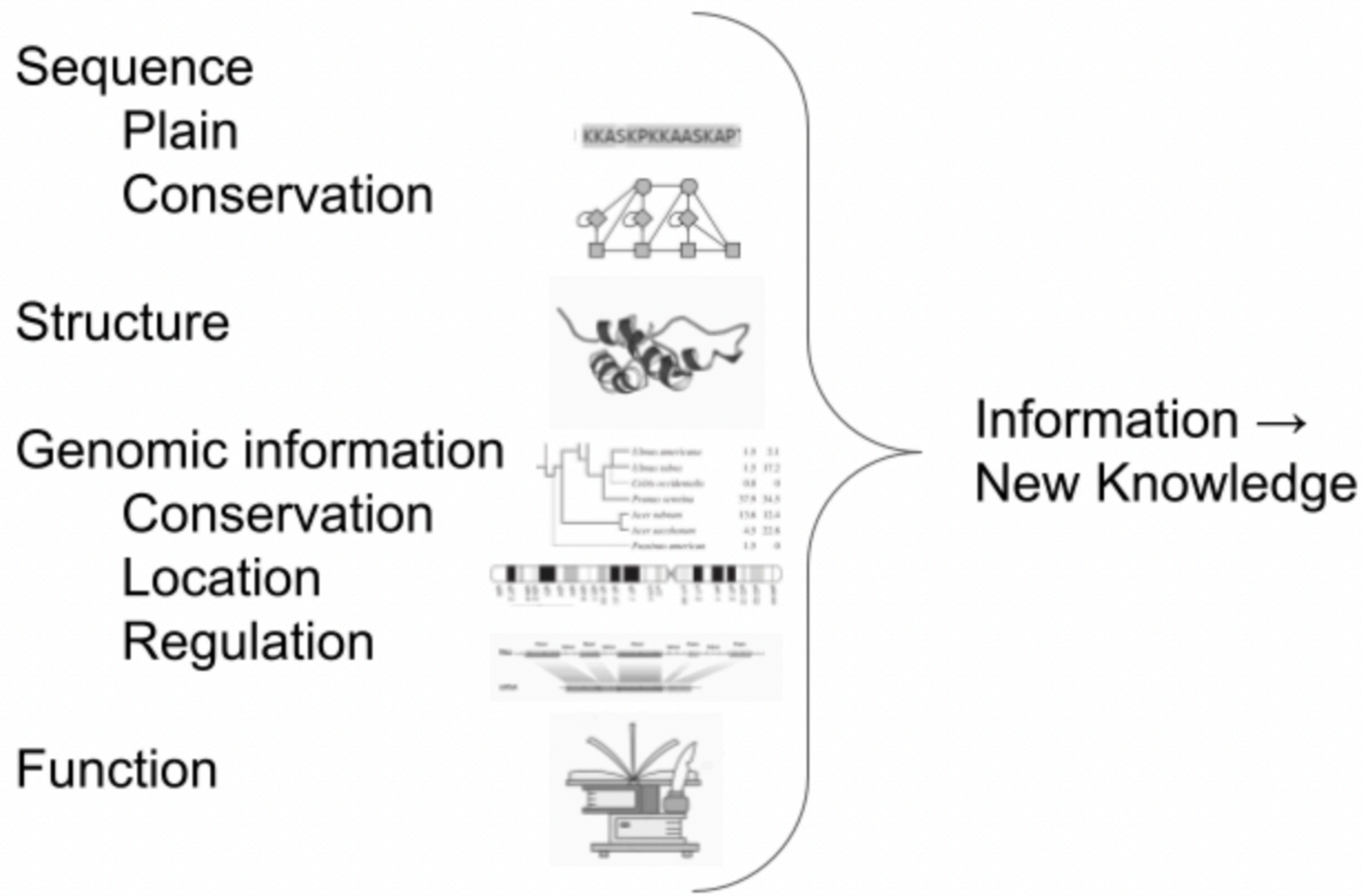
The Bio- in Bioinformatics

Genotype → **Phenotype**

DNA/ Genome ⇒ RNA ⇒ Proteins ⇒ Molecular Networks ⇒ Cells ⇒ Physiology/ Disease

Biological knowledge

Understanding about biological entities comes from crossing the information from/to these different resources and formats



An intricate web of information exists around biological entities, and understanding them involves merging insights from various resources. A big part of some bioinformaticians’ job is to integrate information from different databases and formats to gain a holistic understanding of biological entities.

Features of biological databases

- Data heterogeneity
- High volume of data
- Large scale data integration
- Data sharing / user visualisation and navigation
- Uncertainty / data quality measure needed
- Dynamic and subject to change

Possible classifications of biological databases

Data type

- Genome database
- Sequence database
- Structure database
- Pathway database
- Disease database
- …

Data type

- Genome database
- Sequence database
- Structure database
- Pathway database
- Disease database
- …

Data source

- Primary databases (GenBank, PDB)
- Secondary databases: analysed/aggregated results of the primary ones (UniProtKB)
- Composite database: non-redundant / filtered data (SwissProt)
- …

Biological file formats

- Sequence formats
- Alignment formats
- Features/annotations formats
- Structure formats

Sequence formats

FASTA

File extensions: file.fa, file.fasta, file.fsa

Example:

```
>XR_002086427.1 Candida albicans SC5314 uncharacterized ncRNA (SCR1), ncRNA

TGGCTGTGATGGCTTTTAGCGGAAGCGCGCTGTTCGCGTACCTGCTGTTTGTTGAAAATTTAAGAGCAAAGTGTCCGGCTCGATCCCTGCGAATTGAATTCTGAACGCTAGAGTAATCAGTGTCTTTCAAGTTCT
```

Fasta format is a simple way of representing nucleotide or amino acid sequences of nucleic acids and proteins. This is a very basic format with two minimum lines. First line referred as comment line starts with '>' and gives basic information about sequence. There is no set format for comment line. Any other line that starts with ';' will be ignored. Lines with ';' are not a common feature of fasta files. After comment line, sequence of nucleic acid or protein is included in standard one letter code. Any tabulators, spaces, asterisks etc in sequence will be ignored.

Sequence formats

FASTQ

File extensions: ile.fastq, file.sanfastq, file.fq

Example:

```
@K00188:208:HFLNGBBXX:3:1101:1428:1508 2:N:0:CTTGTA
ATAATAGGATCCCTTTTCCTGGAGCTGCCTTTAGGTAATGTAGTATCTNATNGACTGNCNCCANANGGCTAAAGT
+
AAAFFJJJJJJJJJJJJJJJJJFJJFJJJJJFJJJJJJJJJJJJJJJJ#FJ#JJJJF#F#FJJ#F#JJJFJJJJJ
```

Fastq format was developed by Sanger institute in order to group together sequence and its quality scores (Q: phred quality score). In fastq files each entry is associated with 4 lines.

- Line 1 begins with a '@' character and is a sequence identifier and an optional description.
- Line 2 Sequence in standard one letter code.
- Line 3 begins with a '+' character and is optionally followed by the same sequence identifier (and any additional description) again.
- Line 4 encodes the quality values for the sequence in Line 2, and must contain the same number of symbols as letters in the sequence.

Alignment formats

SAM (Sequence Alignment Map)

File extensions: file.sam

Example:

```
1:497:R:-272+13M17D24M  113     1       497     37      37M     15      100338662       0       CGGGTCTGACCTGAGGAGAACTGTGCTCCGCCTTCAG
19:20389:F:275+18M2D19M 99      1       17644   0       37M     =       17919   314     TATGACTGCTAATAATACCTACACATGTTAGAACCAT   >>>>>>
19:20389:F:275+18M2D19M 147     1       17919   0       18M2D19M        =       17644   -314    GTAGTACCAACTGTAAGTCCTTATCTTCATACTTTGT
9:21597+10M2I25M:R:-209 83      1       21678   0       8M2I27M =       21469   -244    CACCACATCACATATACCAAGCCTGGCTGTGTCTTCT   <;9<<5
```

The SAM Format is a text format for storing sequence data in a series of tab delimited ASCII columns. Most often it is generated as a human readable version of its sister BAM format, which stores the same data in a compressed, indexed, binary form.

SAM format files are generated following mapping of the reads to reference sequence. It is TAB-delimited text format with header and a body. Header lines start with '@' while alignment lines do not. Header hold generic information on SAM file along with version information, if the file is sorted, information on reference sequence, etc. The alignment records constitute the body of the file. Each alignment line/record has 11 mandatory fields describing essential alignment information.

Alignment formats

BAM (Binary Alignment/Map)

File extensions: file.bam

A BAM file is the compressed binary version of the Sequence Alignment/Map (SAM).

a compact and indexable representation of nucleotide sequence alignments. The data between SAM and BAM is exactly same. Being Binary BAM files are small in size and ideal to store alignment files. Require samtools to view the file.

Features/annotations formats

VCF (Variant Calling Format/File)

File extensions: file.vcf

Example:

```
##fileformat=VCFv4.2
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
...
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20 1234567 microsat1 GTC G,GTCT 50 PASS NS=3;DP=9;AA=G GT:GQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```

VCF is a text file format with a header (information VCF version, sample etc) and data lines constitute the body of file.

Features/annotations formats

GFF (General Feature Format or Gene Finding Format)

File extensions: file.gff2, file. gff3, file.gff

Example (GFF2):

```
browser position chr22:10000000-10025000

browser hide all

track name=regulatory description="TeleGene(tm) Regulatory Regions"

visibility=2

chr22 TeleGene enhancer 10000000 10001000 500 + . touch1

chr22 TeleGene promoter 10010000 10010100 900 + . touch1

chr22 TeleGene promoter 10020000 10025000 800 - . touch2
```

GFF (General Feature Format or Gene Finding Format). GFF can be used for any kind of feature (Transcripts, exon, intron, promoter, 3' UTR, repeatitive elements etc) associated with the sequence, whereas GTF is primarily for genes/transcripts. GFF3 is the latest version and an improvement over GFF2 format. However, many databases are still not equipped to handle GFF3 version. The differences will be explained later in text.

The GFF format has 9 mandatory columns and they are TAB separated.

- Col. 1 Reference Sequence
- Col. 2 Source
- Col. 3 Feature
- Col. 4 Start
- Col. 5 End
- Col. 6 Score
- Col. 7 Strand
- Col. 8 Frame (GFF2 and GTF) or Phase (GFF3)
- Col. 9 Attribute or Group field

Features/annotations formats

BED (Browser Extensible Data)

The BED (Browser Extensible Data) file format includes information about sequences that can be visualized in a genome browser; a feature called an annotation track. BED files are tabs-delimited and include 12 fields (columns) of data.

Example of fields: name of chromosome or scaffold, starting position in the chromosome, the ending position…

PSI-MI

The PSI MI format is a data exchange format for molecular interactions.

Example of fields: interaction detection method, biological role, experimental features, location of the interaction, …

PED

File extensions: file.ped

PED is a file format for pedigree analysis, which creates a familial relationship between different samples.

Structure formats

PDB (Protein Data Bank formats)

File extensions: file.pdb

PDB file formats contain atomic coordinates and are used for storing 3D protein structures by the Protein Data Bank.

Example:

```
COMPND    UNNAMED
AUTHOR    GENERATED BY OPEN BABEL 2.3.2
ATOM      1  N   ALA A   1       0.000   0.000   0.000  1.00  0.00           N
ATOM      2  CA  ALA A   1       1.456   0.000   0.000  1.00  0.00           C
ATOM      3  C   ALA A   1       1.930   0.000   1.463  1.00  0.00           C
ATOM      4  O   ALA A   1       1.160   0.000   2.421  1.00  0.00           O
...
CONECT  101   98
CONECT  102   94  103
CONECT  103  102
MASTER        0    0    0    0    0    0    0    0  103    0  103    0
END
```

Other formats

CSV

CSV (.csv file format) files stands for comma separated value and is a text file, where each line is a row and columns are delimited with a comma. It can store different types of sequencing data and can be opened using common spreadsheet programs.

JSON

JSON (JavaScript Object Notation) is a common file format for many other industries, but is used in a growing number of bioinformatics applications and web resources.

And the list of generic file formats goes on…

Why Are There So Many Different Types?

The many different ways of generating and using biological data have given rise to the diversity previously described. These file formats have their own specific use cases depending on:

- Compatibility with specific software
- Data processing, parsing, and human readability needs
- Efficiency for storage

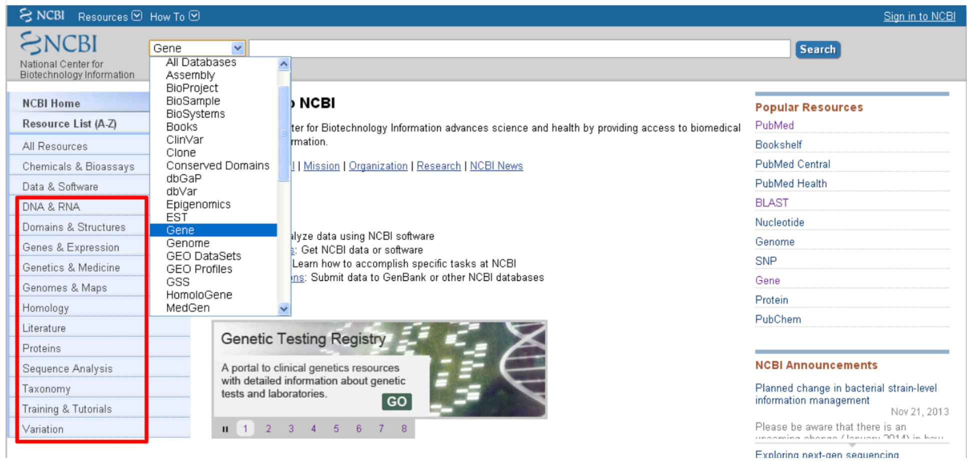
In conclusion, the multitude of biological file formats arises from the diverse needs and characteristics of biological data.

Key Points

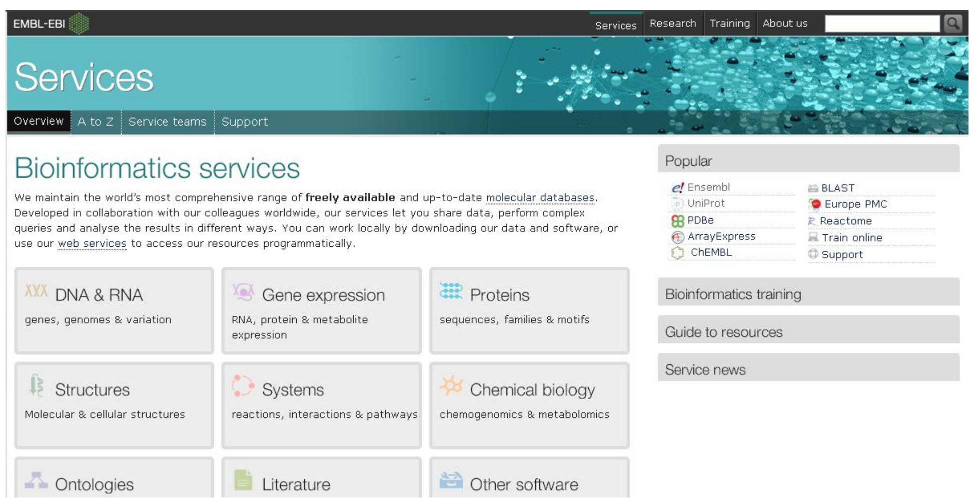
- Biological data is multi-layered. E.g. the information about one gene can actually regard multiple different biological entities: the variability of its sequence, the derived protein, the diseases associated etc.
- Consequently, several different sources of information can be identified and used to describe a biological entity, as well as several different file formats.

Centralized *vs.* individual resources

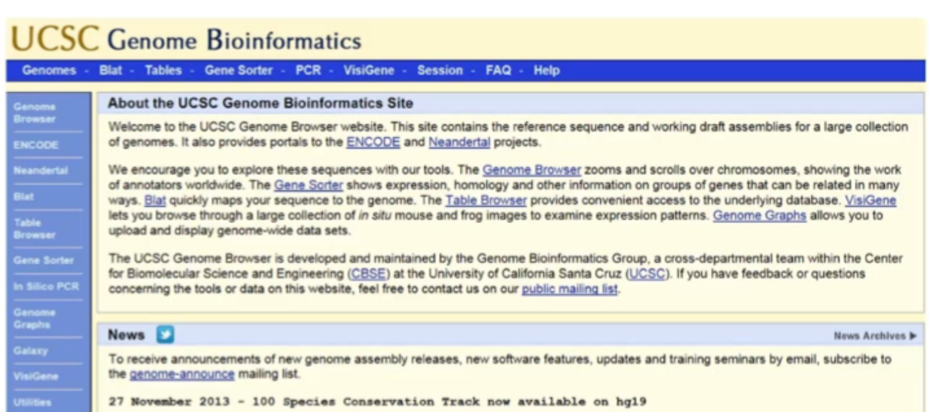
NCBI: http://www.ncbi.nlm.nih.gov/

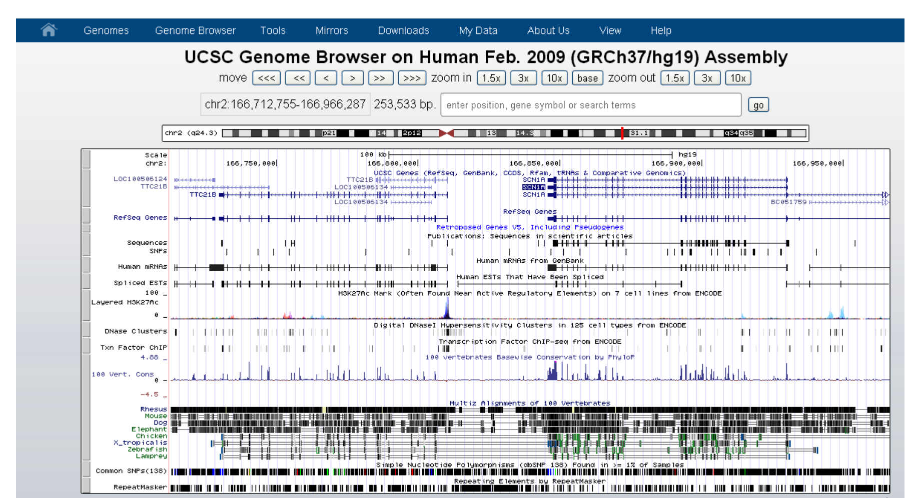


EBI: http://www.ebi.ac.uk/services



UCSC Genome Bioinformatics (http://genome.ucsc.edu/)





NCBI/EBI/UCSC Summary Table

	NCBI	EBI	UCSC
Tools	BLAST	BLAST Exonerate ClustalW2	BLAT In-Silico PCR
Data Repository	GenBank GEO SRA	ArrayExpress ENA PDBe	ENCODE
DNA/Genome	Genome	Ensembl	Ideogram Recombination Rate GC Content

NCBI/EBI/UCSC Summary Table (Cont'd)

	NCBI	EBI	UCSC
DNA/Gene	Gene	Ensembl	UCSC Genes GENCODE RefSeq Genes
RNA	RefSeq	Ensembl	mRNAs ESTs UniGene
Proteins	Protein, RefSeq Conserved Domain	UniProt, InterPro, PRIDE (mass spec)	

NCBI/EBI/UCSC Summary Table (Cont'd)

	NCBI	EBI	UCSC
Expression	UniGene	Expression Atlas	Affy Exon Array Caltech RNA-seq Allen Brain
Regulation			Transcription TFBS Epigenetics DNaseI HS
Literature	PubMed		
Ontology		Gene Ontology	

NCBI/EBI/UCSC Summary Table (Cont'd)

	NCBI	EBI	UCSC
Comparative Genomics	Taxonomy HomoloGene	*Ensembl	Conservation Neandertal
Variation	dbSNP dbVar	*Ensembl	SNPs DGV RepeatMasker
Disease	OMIM MeSH dbGaP ClinVar		GAD COSMIC ClinVar GWAS Catalog QTLs

NCBI access

• AllResources – Database:

http://www.ncbi.nlm.nih.gov/guide/all/#databases_

– Tools:

http://www.ncbi.nlm.nih.gov/guide/all/#tools_

• Resource List (A-Z)

– http://www.ncbi.nlm.nih.gov/guide/sitemap/

EBI access

- Bioinformatics services

– by topic:

http://www.ebi.ac.uk/services

- DNA&RNA,Geneexpression,Proteins,Structures,Systems, Chemical biology, Ontologies, Literature, ...

- Popular: Ensembl, UniProt, PDBe, ArrayExpress, ChEMBL, BLAST, Europe PMC, Reactome, Train online, ...

– by name(A-Z):

http://www.ebi.ac.uk/services/all

UCSC access

- Table Browser
  - http://genome.ucsc.edu/cgi-bin/hgTables
- GenomeBrowser
  - http://genome.ucsc.edu/cgi-bin/hgTracks • trackhubs

• configure

Examples of individual Resources

Analyses		Databases and tools
DNA/Genome	Gene prediction	GENSCAN, Glimmer
	Genetic and Somatic Variations	HGMD, COSMIC, SIFT, PolyPhen, SAPRED
Expression regulation	Transcription factors & binding	TRANSFAC, PlantTFDB
	RNA annotation	Rfam
	microRNA annotation	miRBase
Epigenetics	DNA methylation	MethylomeDB
Pathway & network	Pathway	KEGG, PANTHER, BioCyc, REACTOME, KOBAS, DAVID
	Interaction network	PID, STRING
Evolution	Conservation	GERP++, PHYML

Examples of individual resources for mass spectrometry proteomics analyses

	Databases and tools
Mass spec data repository	The Global Proteome Machine
Peptide identification by database search	Sequest, Mascot, ProteinProspector, pFind, PEAKS, Byonic, Proteome Discoverer, SpectrumMill, Masslynx, X!Tandem
de novo peptide identification	pNovo, PEAKS, PepNovo
Quantitation	MaxQuant, Census

Examples of Individual Resources for Protein 3D Structures

Analyses	Databases and Tools
Protein 3D structures	PDB
Homology Modeling	Modeller, Swiss-Model, I-TASSER
Fold Recognition	3D-PSSM, Phyre2, PROSPECT
ab inito folding	QUARK, Rosetta
Protein 3D models	Swiss-Model Repository
Nucleotide structure	Mfold, PDB
Nucleotide interaction	RNAhybrid

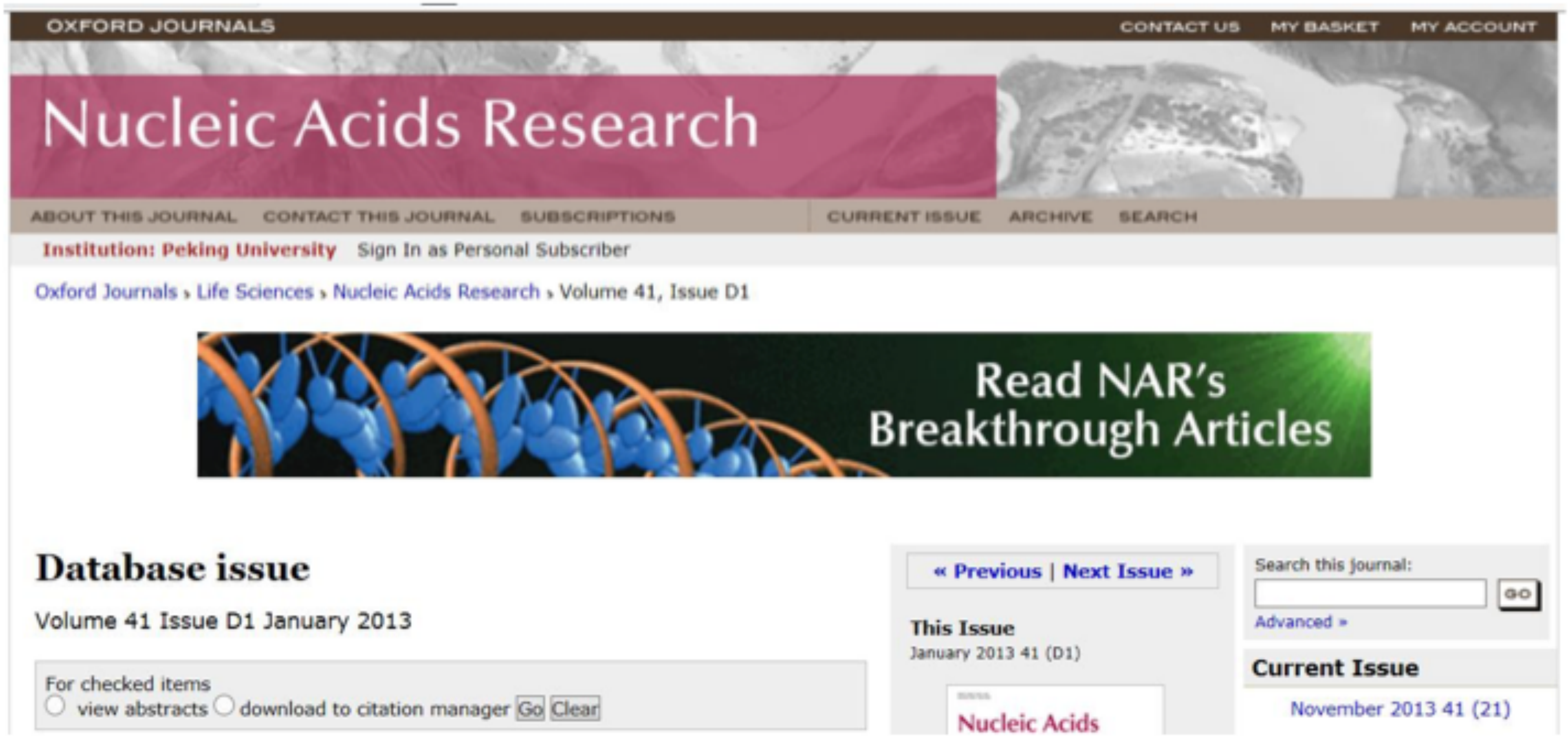
Examples of Individual Resources for NGS analysis

Analyses		Databases and tools
Read mapping	Reads mapping (DNA)	BWA, Bowtie/Bowtie2
	Reads mapping (RNA)	TopHat/TopHat2
	Reads mapping related utilities	GATK, FastQC, RNA-SeQC, SAMtools, Picard
Assembly	de novo genomic assembly	Velvet, SOAP de novo
	de novo transcriptome assembly	Trinity, Velvet + Oases
	reference-based transcriptome assembly	TopHat + Cufflinks
Visualization	Genome Browser	GBrowse, JBrowse, IGV
Variant calling	SNP	GATK, SOAPsnp, SAMtools
	Indel	Pindel
	CNV and structural variation	CNVnator, BIC-seq, SVMerge, mrCaNaVaR, ExomeCNV, CoNIFER, HMMcopy, Control-FREEC
Expression	Differential expression	Cuffdiff, DESeq/DESeq2
ChIP-Seq	Peak calling	MACS

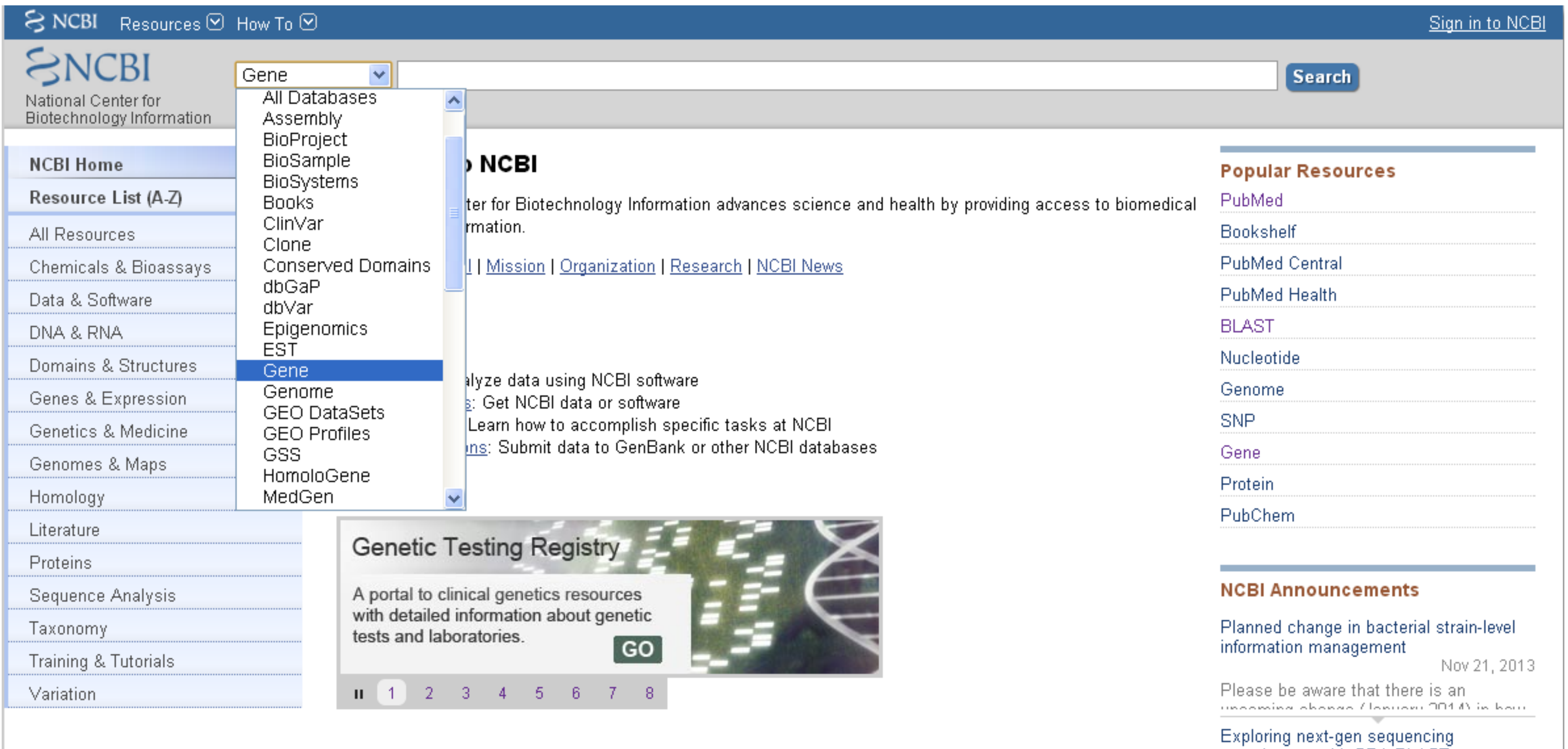
Other examples of individual resources & utilities

Purpose		Databases and tools
Model Organisms	Genome and gene annotations	Flybase, Wormbase, ZFIN, TAIR
Large-scale studies	Cancer	TCGA, CGP
	Epigenetics	Roadmap Epigenomics Project
	Brain	Allen Brain Atlas, Human Connectome project
Tools to assist wet-lab experiments	Primer design	Primer3/Primer3Plus, Electronic PCR
Software programming utilities	Sequence analysis tool package	EMBOSS
	R package	Bioconductor
	Perl package	BioPerl
	Python package	BioPython
Workflow	Workflow platform	Galaxy
	Workflow construction	Taverna

NAR database issue & web server issue



NCBI: http://www.ncbi.nlm.nih.gov/

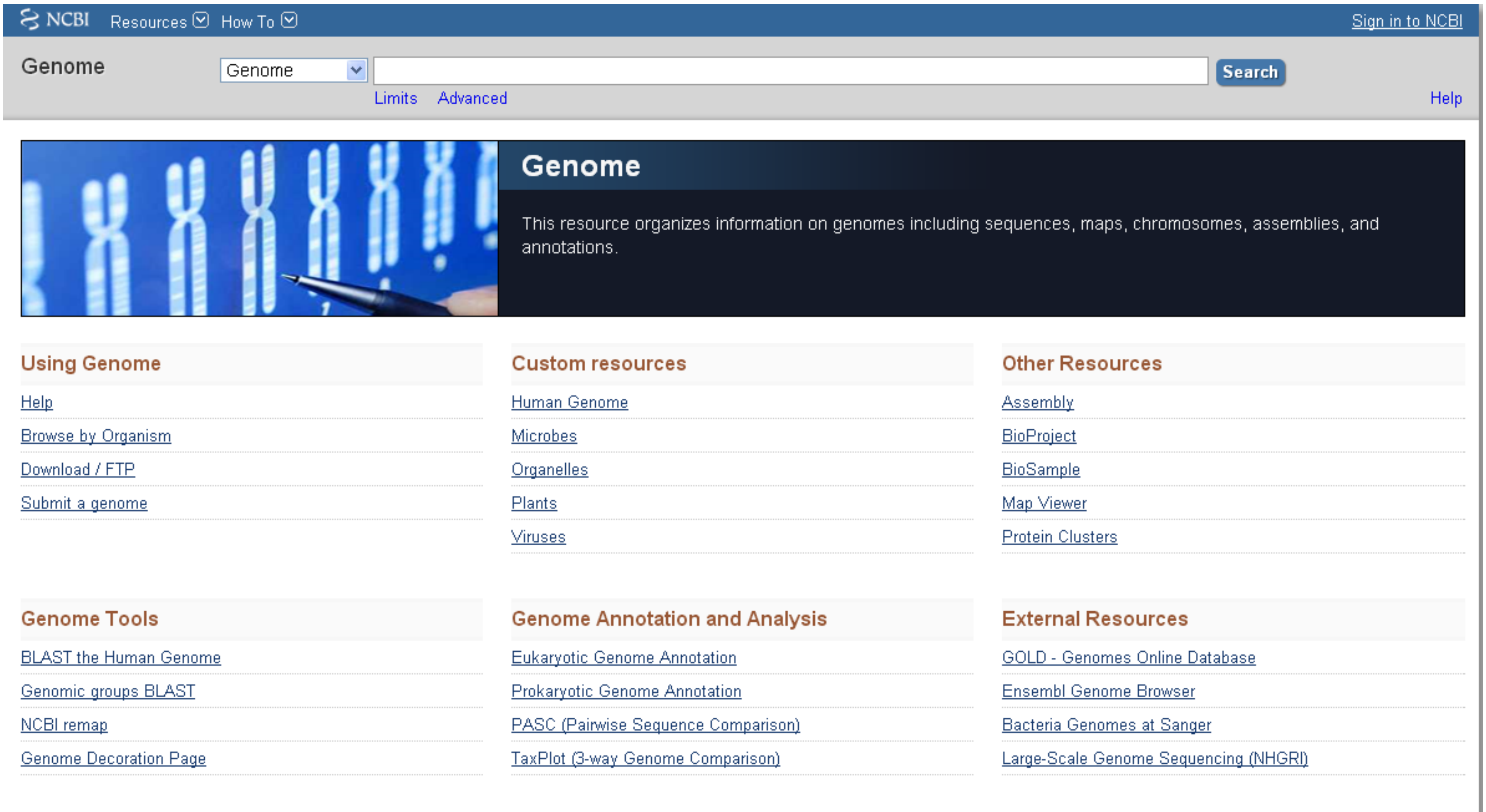


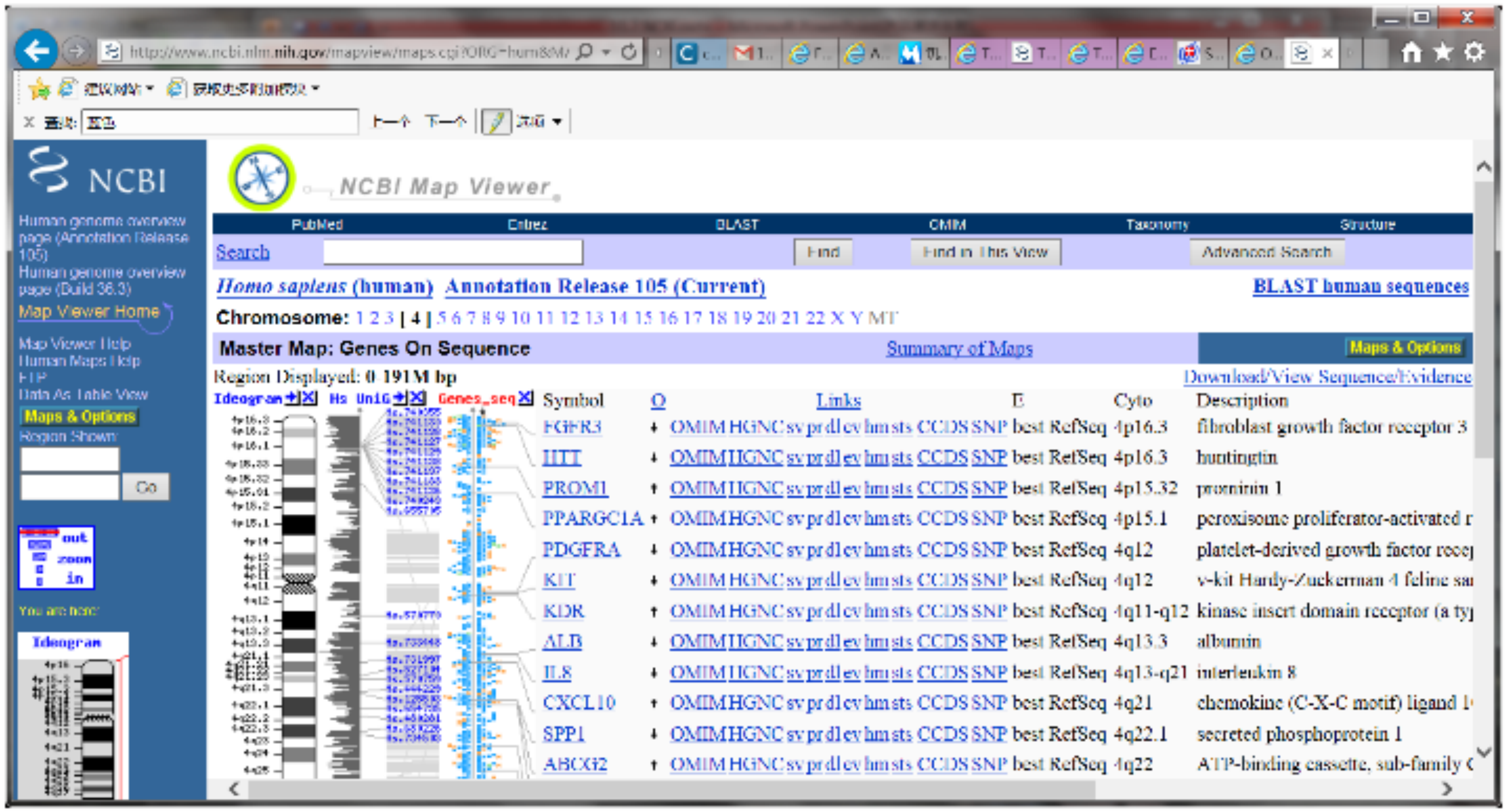
Examples of resources at NCBI

Data Repository	GenBank, dbEST, GEO, SRA
DNA	Genome, Gene
Comparative Genomics	Taxonomy, HomoloGene
Genetic Variation	dbSNP, dbVar
Disease Mutations	OMIM, dbGaP, ClinVar
RNA	RefSeq, UniGene
Proteins	Protein, RefSeq, Conserved Domain
Literature	PubMed, MeSH
Tools	BLAST

NCBI-Genome

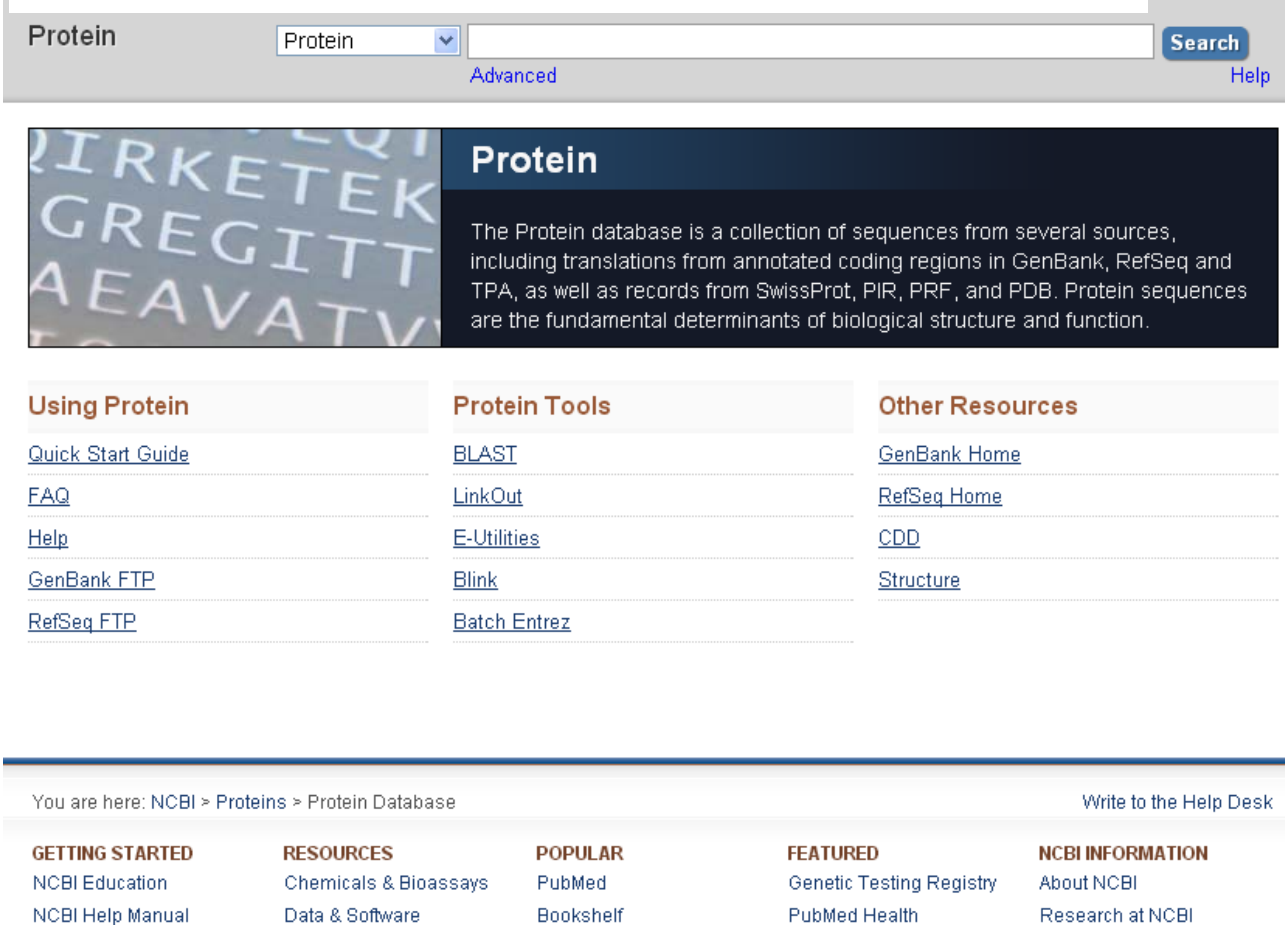
http://www.ncbi.nlm.nih.gov/genome/




http://www.ncbi.nlm.nih.gov/mapview/maps.cgi?ORG=hum&MAP
NCBI
Human genome overview page (Annotation Release 105)
Human genome overview page (Build 36.3)
Map Viewer Home
Map Viewer Help
Human Maps Help
FTP
Data As Table View
Maps & Options
Region Shown:
Go
out
zoom
in
You are here:
Ideogram
NCBI Map Viewer
PubMed
Entrez
BLAST
OMIM
Taxonomy
Structure
Search
Find
Find in This View
Advanced Search
Homo sapiens (human) Annotation Release 105 (Current)
BLAST human sequences
Chromosome: 1 2 3 | 4 | 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 X Y MT
Master Map: Genes On Sequence
Summary of Maps
Maps & Options
Region Displayed: 0-191M bp
Download/View Sequence/Evidence
Ideogram
Hs UniG
Genes_seq
Symbol
O
Links
E
Cyto
Description
FGFR3 OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4p16.3 fibroblast growth factor receptor 3
HTT OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4p16.3 huntingtin
PROM1 OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4p15.32 prominin 1
PPARGC1A OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4p15.1 peroxisome proliferator-activated r
PDGFRA OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4q12 platelet-derived growth factor rece
KIT OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4q12 v-kit Hardy-Zuckerman 4 feline sa
KDR OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4q11-q12 kinase insert domain receptor (a ty
ALB OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4q13.3 albumin
IL8 OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4q13-q21 interleukin 8
CXCL10 OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4q21 chemokine (C-X-C motif) ligand 1
SPP1 OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4q22.1 secreted phosphoprotein 1
ABCG2 OMIM HGNC sv pr dl ev hm sts CCDS SNP best RefSeq 4q22 ATP-binding cassette, sub-family G

NCBI-Nucleotide/Protein (RefSeq)

www.ncbi.nlm.nih.gov/nuccore/ www.ncbi.nlm.nih.gov/protein/



NCBI-Nucleotide/Protein (RefSeq): [NM_*, NP_*]

www.ncbi.nlm.nih.gov/nuccore/ www.ncbi.nlm.nih.gov/protein/

transcript variant 1, mRNA

NCBI Reference Sequence: NM_001017424.2

FASTA Graphics

Go to:

```
LOCUS       NM_001017424            3597 bp    mRNA    linear   PRI 17-NOV-2013
DEFINITION  Homo sapiens potassium channel, subfamily K, member 2 (KCNK2),
            transcript variant 1, mRNA.
ACCESSION   NM_001017424
VERSION     NM_001017424.2  GI:126365744
KEYWORDS    RefSeq.
SOURCE      Homo sapiens (human)
  ORGANISM  Homo sapiens
            Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
            Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini;
            Catarrhini; Hominidae; Homo.
REFERENCE   1  (bases 1 to 3597)
```

Customize vi

Analyze this s

Run BLAST

Pick Primers

Highlight Seque

Find in this Seq

Articles abou gene

Regulation of int

n J Physiol Lung

NCBI-Gene

http://www.ncbi.nlm.nih.gov/gene/

Display Settings: Full Report

Send to:

ABCA1 ATP-binding cassette, sub-family A (ABC1), member 1 [*Homo sapiens* (human)]

Gene ID: 19, updated on 21-Nov-2013

Summary

Official Symbol	ABCA1 provided by HGNC
Official Full Name	ATP-binding cassette, sub-family A (ABC1), member 1 provided by HGNC
Primary source	HGNC:29
See related	Ensembl:ENSG00000165029; HPRD:02501; MIM:600046; Vega:OTTHUMG00000020417
Gene type	protein coding
RefSeq status	REVIEWED
Organism	Homo sapiens
Lineage	Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo
Also known as	TGD; ABC1; CERP; ABC-1; HDLDT1
Summary	The membrane-associated protein encoded by this gene is a member of the superfamily of ATP-binding cassette (ABC) transporters. ABC proteins transport various molecules across extra- and intracellular membranes. ABC genes are divided into seven distinct subfamilies (ABC1, MDR/TAP, MRP, ALD, OABP, GCN20, White). This protein is a member of the ABC1 subfamily. Members of the ABC1 subfamily comprise the only major ABC subfamily found exclusively in multicellular eukaryotes. With cholesterol as its substrate, this protein functions as a cholesteral efflux pump in the cellular lipid removal pathway. Mutations in this gene have been associated with Tangier's disease and familial high-density lipoprotein deficiency. [provided by RefSeq, Jul 2008]

Genomic context

Location: 9q31.1

Sequence: Chromosome: 9; NC_000009.11 (107543283..107690527, complement)

See ABCA1 in Epigenomics, MapViewer

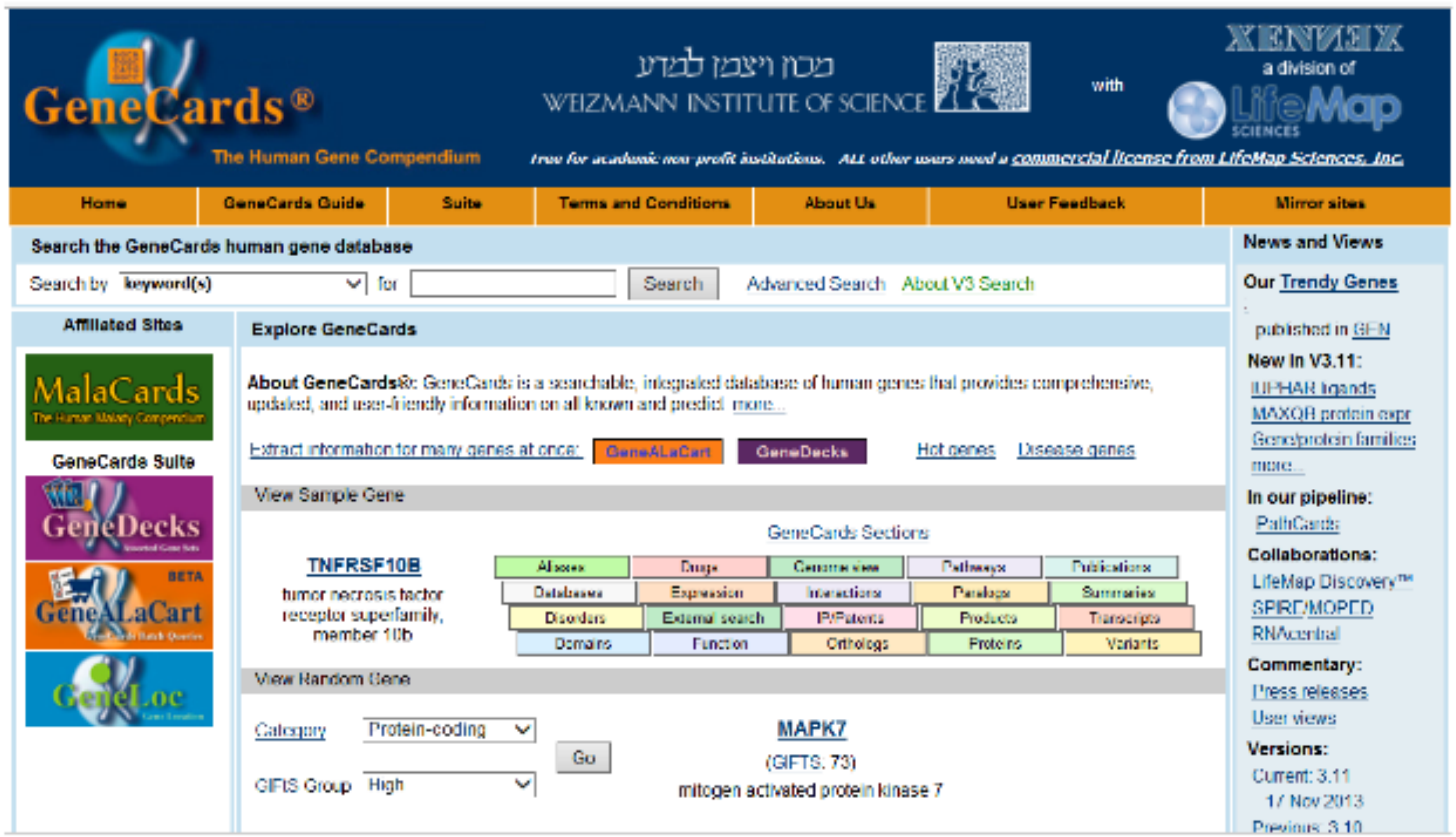
Chromosome 9 - NC_000009.11

107509969 108200785

Related information

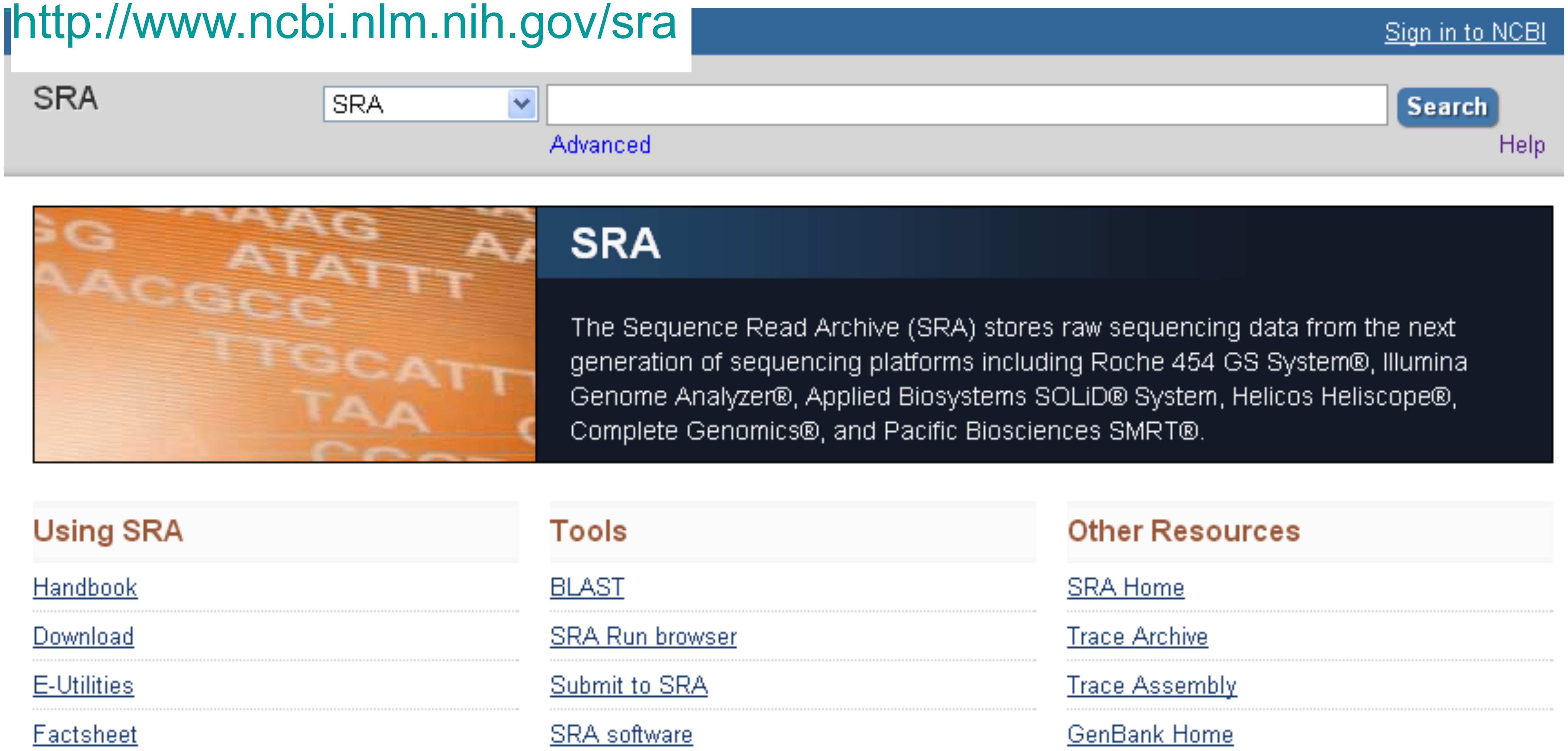
- Order cDNA clone
- BioAssay
- BioAssay by Target (List)
- BioAssay by Target (Summary)
- BioProjects
- BioSystems
- Books

Human genes: GeneCards http://www.genecards.org/



NCBI-SRA

http://www.ncbi.nlm.nih.gov/sra



NCBI-SRA

http://www.ncbi.nlm.nih.gov/sra

ERX033410: Foxa1 and Foxa2 are essential for gender dimorphism in liver cancer
1 ILLUMINA (Illumina Genome Analyzer IIx) run: 40.9M spots, 1.5G

Accession: ERX033410
Experiment design: Foxa1 and Foxa2 are essential for gender dim
Submission: ERA070841 by UNIVERSITY OF PENNSYLVANIA
Study summary: Foxa1 and Foxa2 are essential for gender dimorpl
Study • All experiments (more...)
Sample: Treated-Male-Control (ERS074997) (less...)
Organism: Mus musculus
Attributes:
OrganismPart: liver
Genotype: Foxa1loxP/loxP;Foxa2loxP/loxP
StrainOrLine: 129J/BL6
Organism: Mus musculus
Sex: male
Library: MDAR (more...)
Platform: Illumina (less...)
Instrument model: Illumina Genome Analyzer IIx
Spot descriptor:
forward
1

Experiment attributes:
Experimental Factor: ORGANISM: Mus musculus
Experimental Factor: SEX: male
Experimental Factor: TREATMENT: Treated
Experimental Factor: GENOTYPE: Foxa1loxP/loxP;Foxa2loxP/loxP
Experimental Factor: IMMUNOPRECIPITATE: AR
Total: 1 run, 40.9M spots, 1.5G bases, 918.8Mb

#	Run	# of Spots	# of Bases	Size
1.	ERR056365	40,948,224	1.5G	918.8Mb

ID: 197484

Change accessi

Run ERR056365 (Foxa1 and Foxa2 are essential for gender dimorphism in liver cancer)

Metadata | Reads | Download

Run	Spots	Bases	Size	GC content	Published	Access Type
ERR056365	40.9 M	1.5 Gbp	963.4 M	43.8%	2012-06-26	public

Quality graph (bigger)

This run has 1 read per spot:

L=36, 100%

Legend

Experiment	Library					
	Name	Platform	Strategy	Source	Selection	Layout
ERX033410	MDAR	Illumina	ChIP-Seq	GENOMIC	ChIP	SINGLE

Show design

Biosample	Sample name	Title
ERS074997	E-MTAB-805:Treated-Male-Control	Treated-Male-Control

Bioproject	SRA Study	Title
	ERP001029	Foxa1 and Foxa2 are essential for gender dimorphism in liver cancer

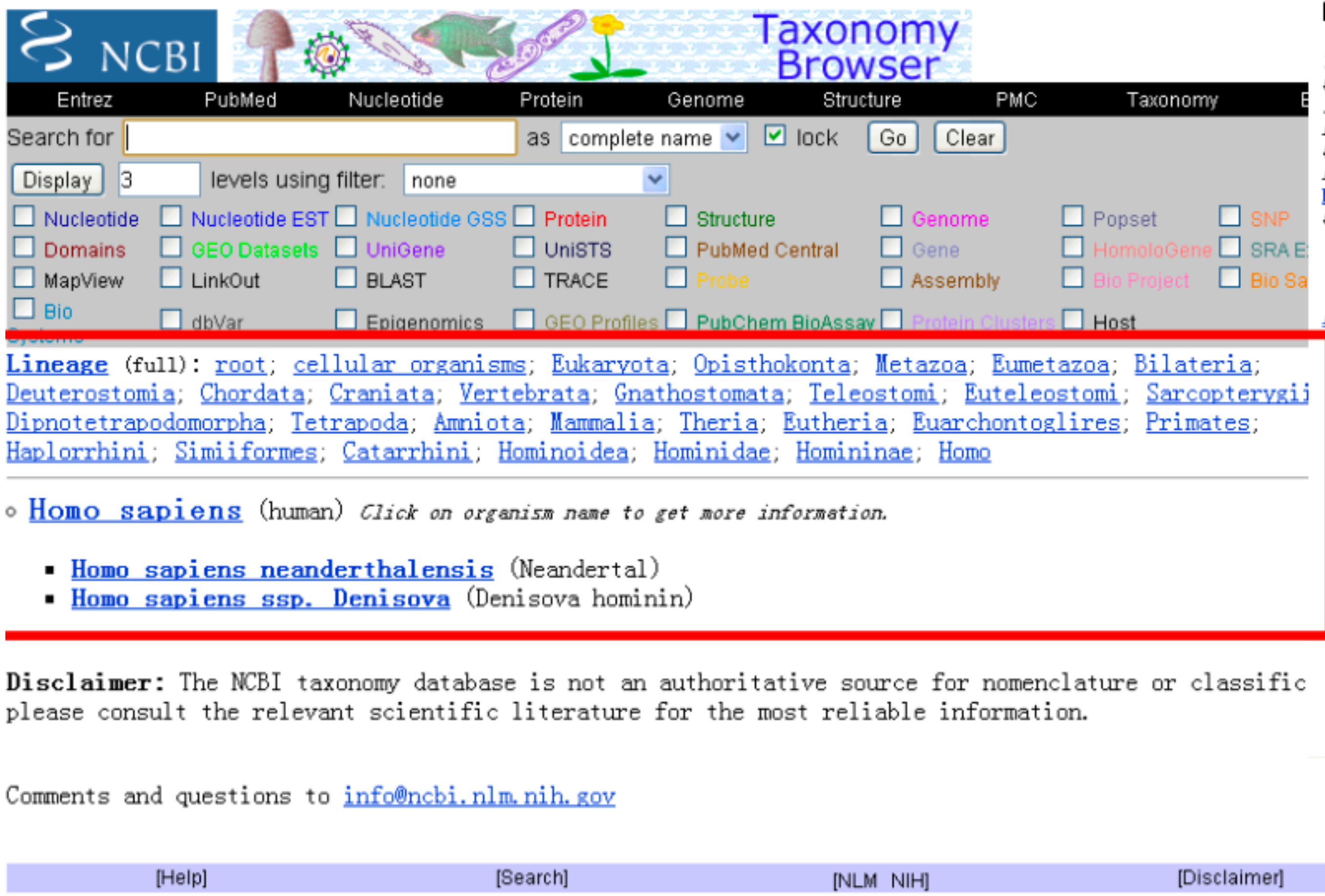
Metadata | Reads | **Download**

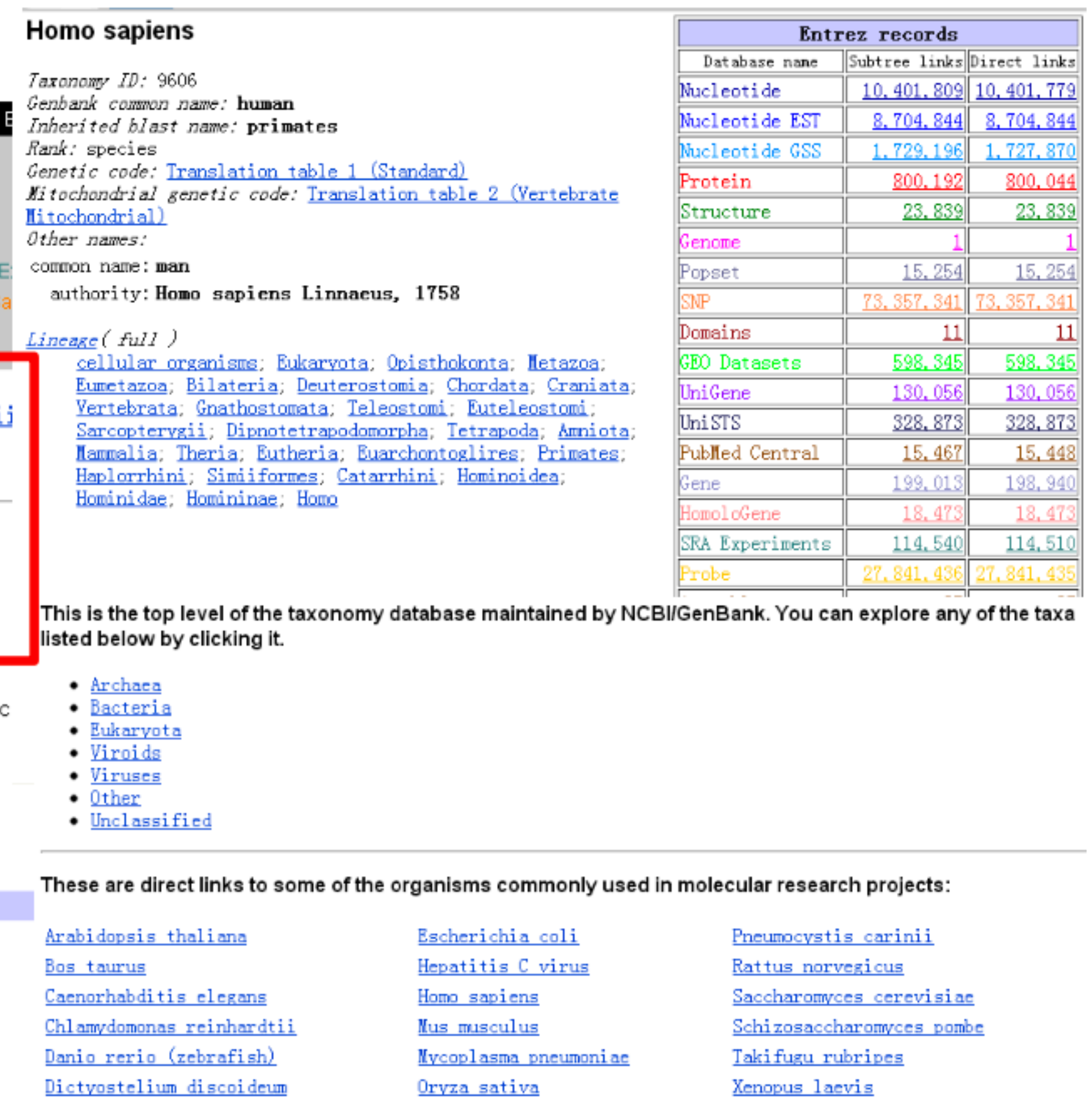
Object		.sra		
Run	ERR056365	963.4 Mb	HTTP	FTP
Experiment	ERX033410	963.4 Mb	HTTP	FTP
Study	ERP001029	12.0 Gb	HTTP	FTP

(use Aspera plugin for fast download)

NCBI-Taxonomy

http://www.ncbi.nlm.nih.gov/taxonomy/





NCBI-Taxonomy Statistics

http://www.ncbi.nlm.nih.gov/taxonomy/

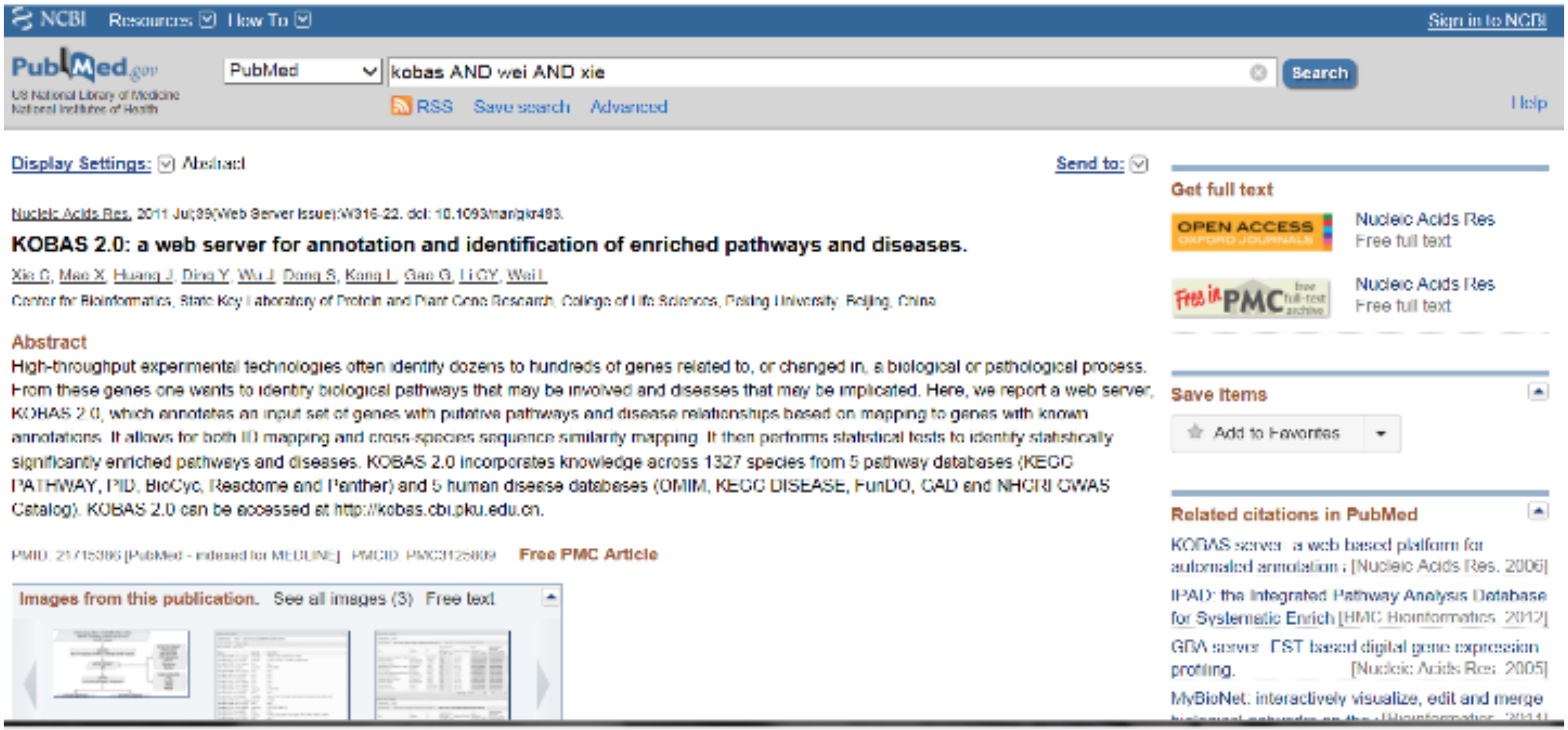
Taxonomy Nodes (all dates)

Ranks:	higher taxa	genus	species	lower taxa	total
Archaea	703	291	970	0	1,964
Bacteria	6,769	5,257	26,278	964	39,268
Eukaryota	69,307	100,515	539,227	38,309	747,358
Fungi	6,335	7,688	59,812	1,584	75,419
Metazoa	50,074	71,780	280,733	19,250	421,837
Viridiplantae	8,685	17,003	183,434	17,080	226,202
Viruses	2,258	2,794	5,976	793	11,821
All taxa	79,068	108,858	572,437	40,066	800,429

Dates: 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024 all dates

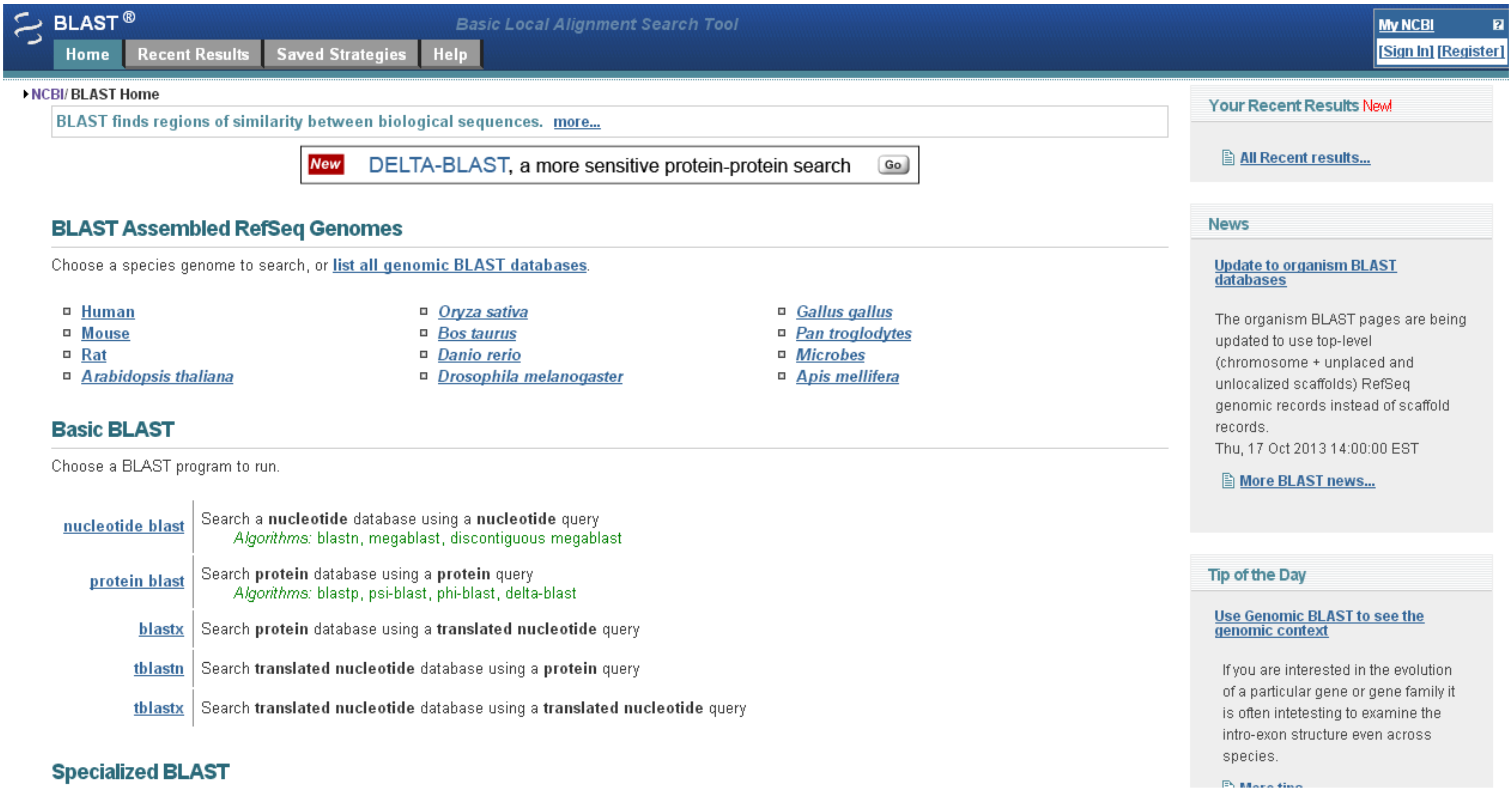
NCBI-PubMed

http://www.ncbi.nlm.nih.gov/pubmed/



NCBI-BLAST

http://blast.ncbi.nlm.nih.gov/Blast.cgi



NCBI-BLAST

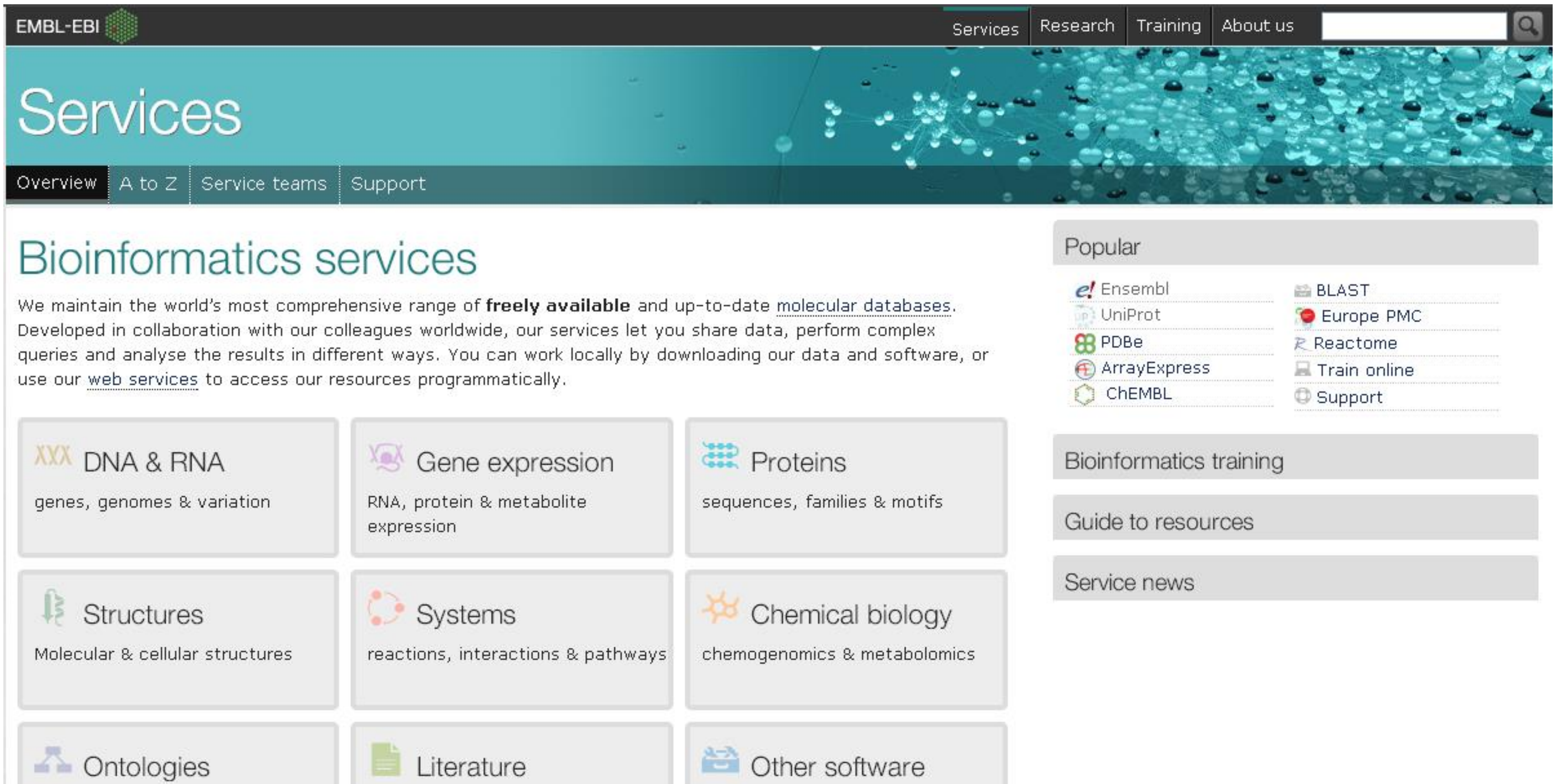
http://blast.ncbi.nlm.nih.gov/Blast.cgi

– Online: NCBI-BLAST(blast.ncbi.nlm.nih.gov/Blast.cgi)

– Standalone: BLAST+

– Embedded in webpage: wwwblast

EBI (http://www.ebi.ac.uk/services)



DNA, RNA, and proteins

Genes and Genomes

Databases	
Comprehensive annotation	**Ensembl**
Sequence archive	**ENA, EGA, EBI Metagenomics**
Large-scale sequencing project	**1000 Genomes**
Tools	
Sequence alignment	**BLAST, Clustal Omega**

Expression

Databases	
Expression profile archive	**ArrayExpress, Expression Atlas**

Proteins

Databases	
Protein annotation	**UniProt, InterPro, Pfam**
Proteomics data archive	**PRIDE**
Protein structure	**PDBe, PDBsum, Quaternary structure**
Tools	
Protein function prediction	**InterProScan**
Protein structure prediction	**PDBePISA**
Protein function prediction	**ProFunc**
Protein structure alignment	**PDBeFold**

Reactions, pathways, ontologies

Interactions, pathways & networks

Databases	
Molecular interaction	IntAct
Pathway database	Reactome
Network dynamics	BioModels

Small molecules

Databases	
Annotations of chemicals	ChEBI, ChEMBL
Metabolomics archive	Metabolights

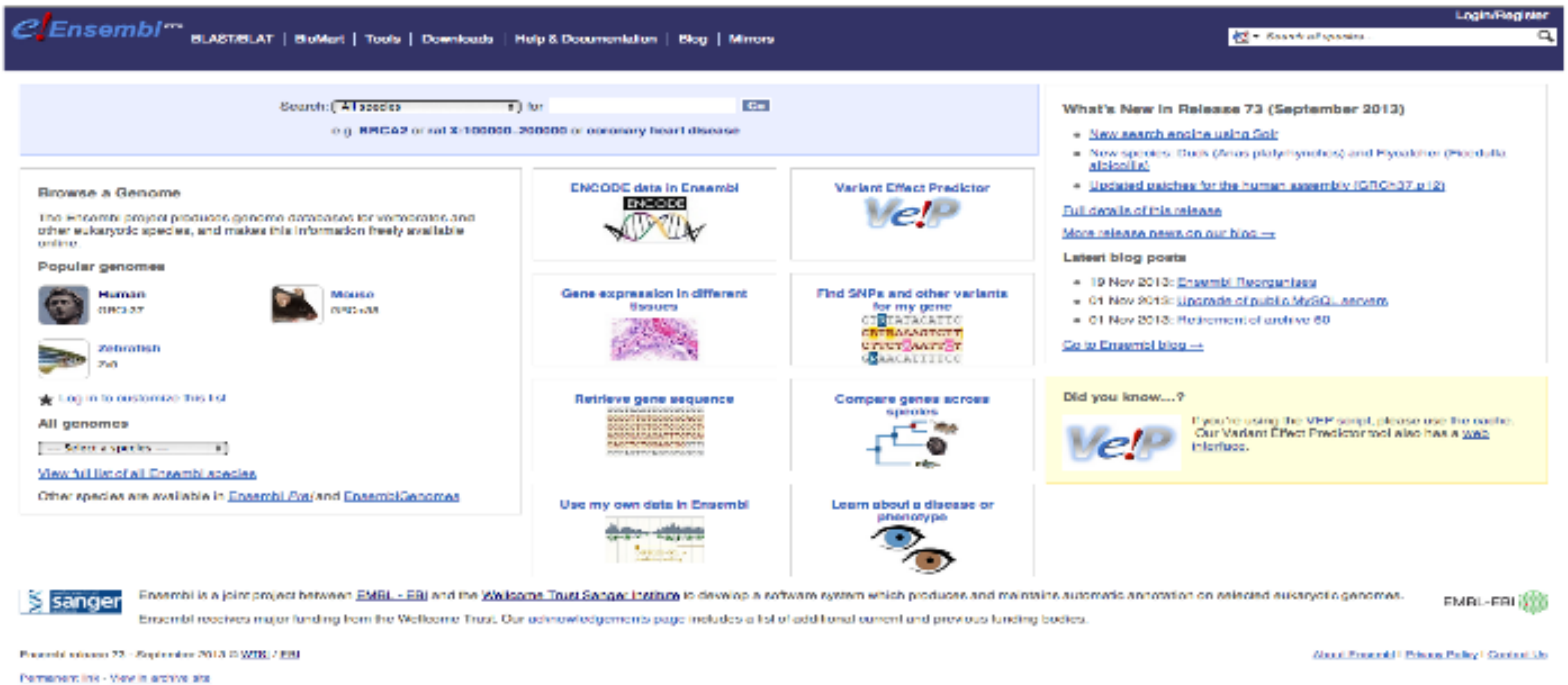
Enzymes and reactions

Databases	
Enzyme annotation	EBI Enzyme Portal, Catalytic Site Atlas

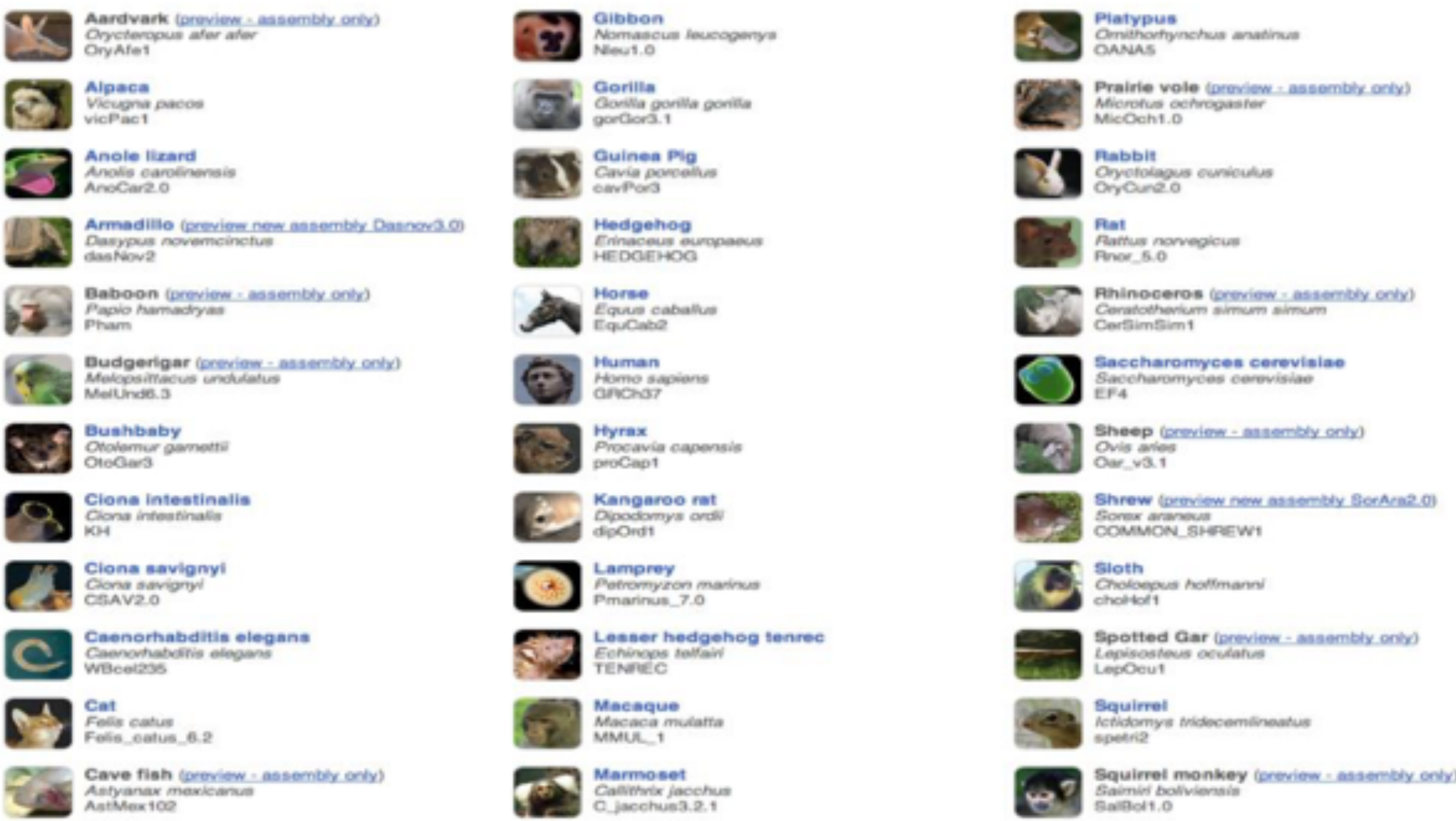
Ontologies

Databases	
Ontologies	Gene Ontology, Systems Biology Ontology, Experimental Factor Ontology
Tools	
Search engine for ontologies	Ontology Lookup Service, QuickGO

Ensembl (http://ensembl.org)



Lots of Species in Ensembl



Four Types of Data in Ensembl

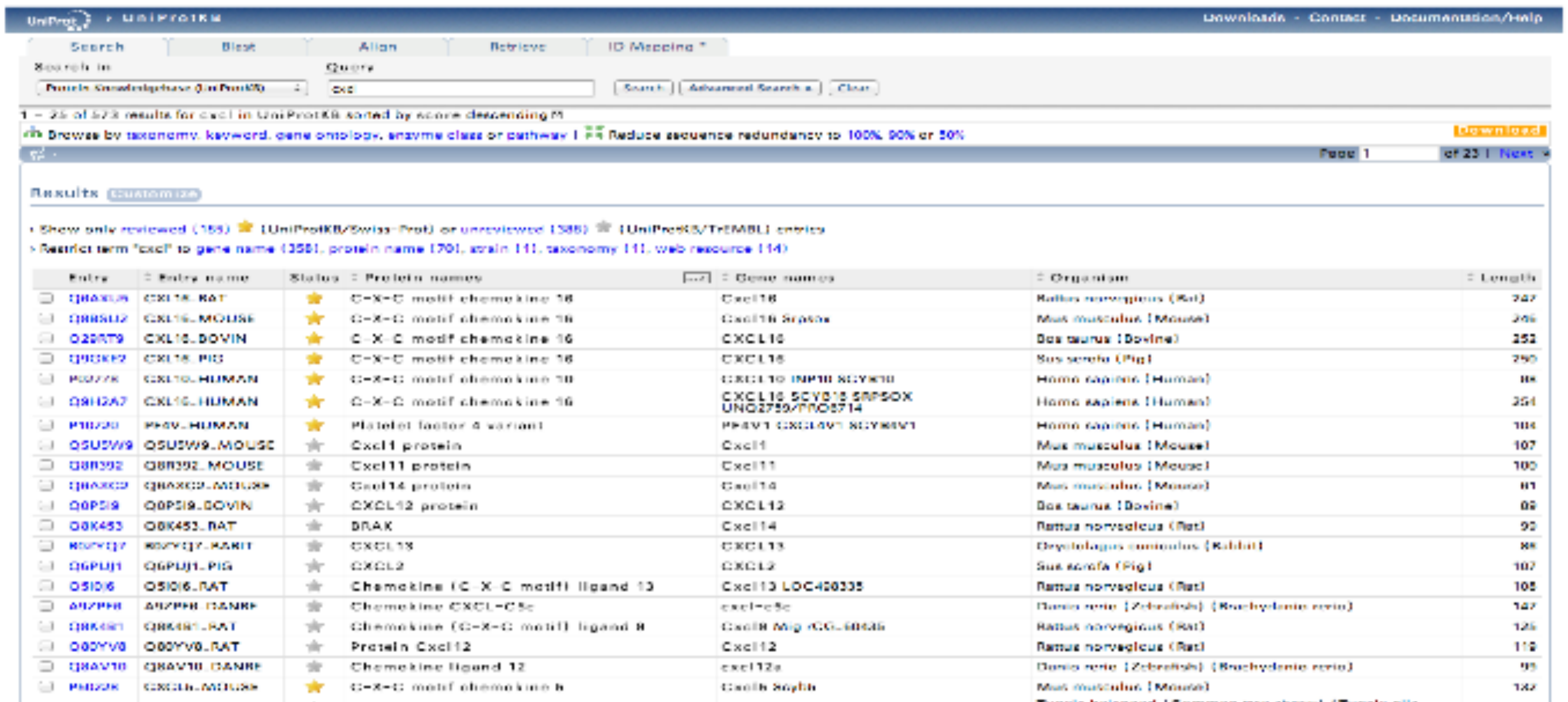
- **Curated/Reference**: Data that are manually curated based on experimental results and validations. They serve as the “reference” for future studies.

- **Large-scale projects**: Data that are not curated but are produced by high-throughput experiments. They tend to have a large sample size or data volume.

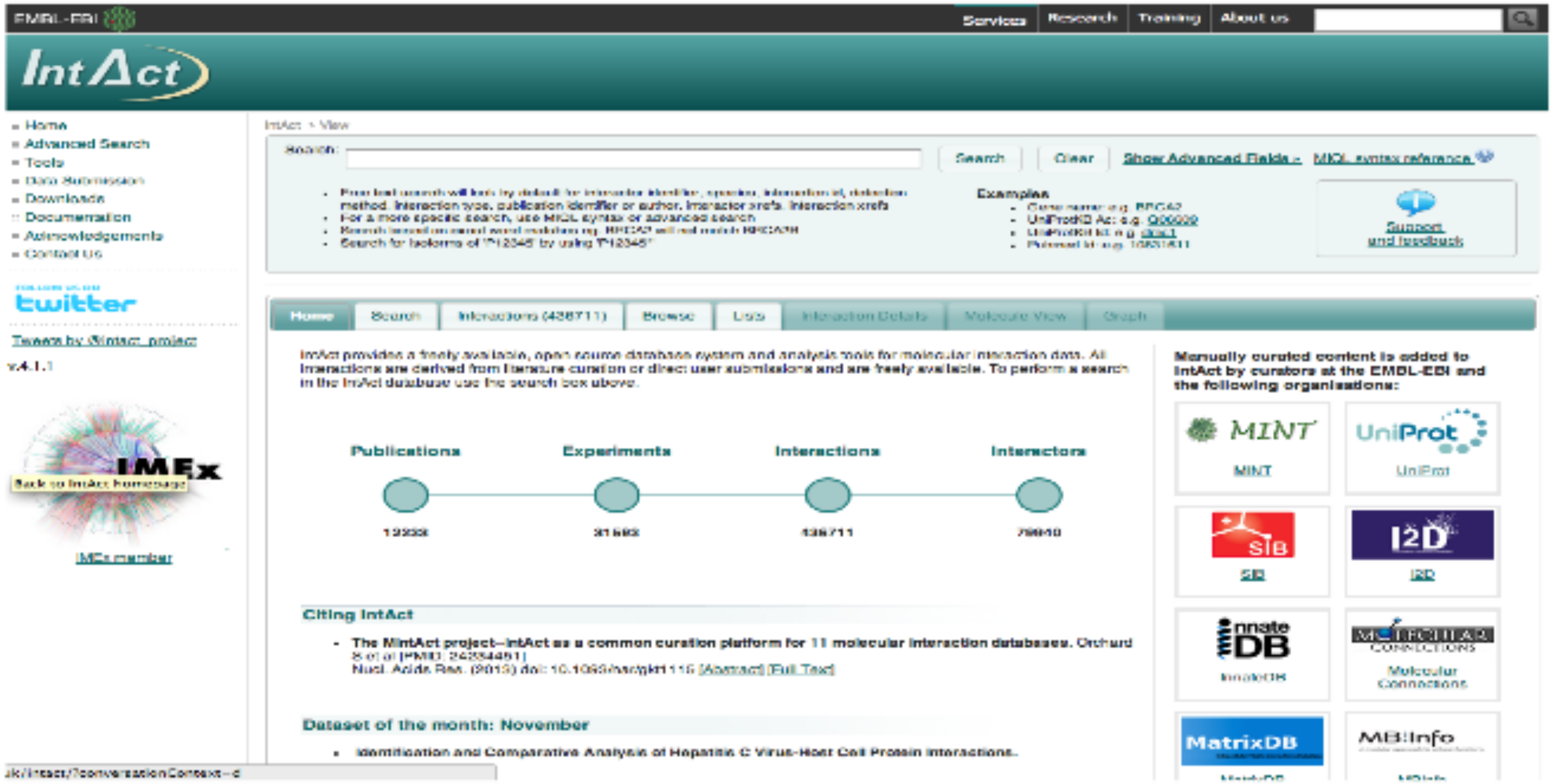
- **Other studies**: Low-throughput experimental data that are not curated, possibly with some computational postprocessing.

- **Only computationally analyzed**: Secondary data that are generated by running automated algorithms over curated/reference data and large-scale high-throughput data (e.g. the genome sequence).

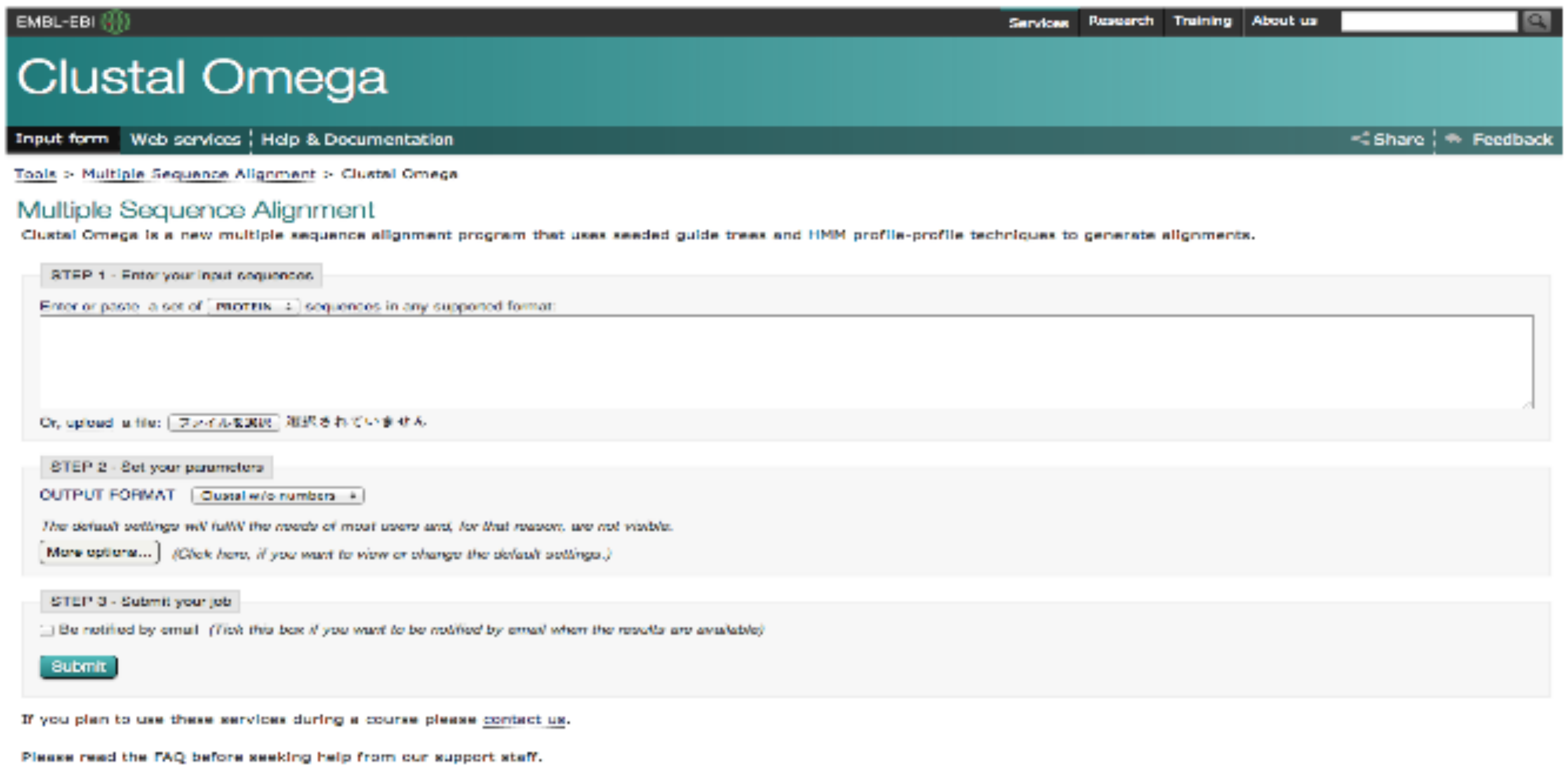
UniProtKB (http://www.uniprot.org/)



IntAct (http://www.ebi.ac.uk/intact/)

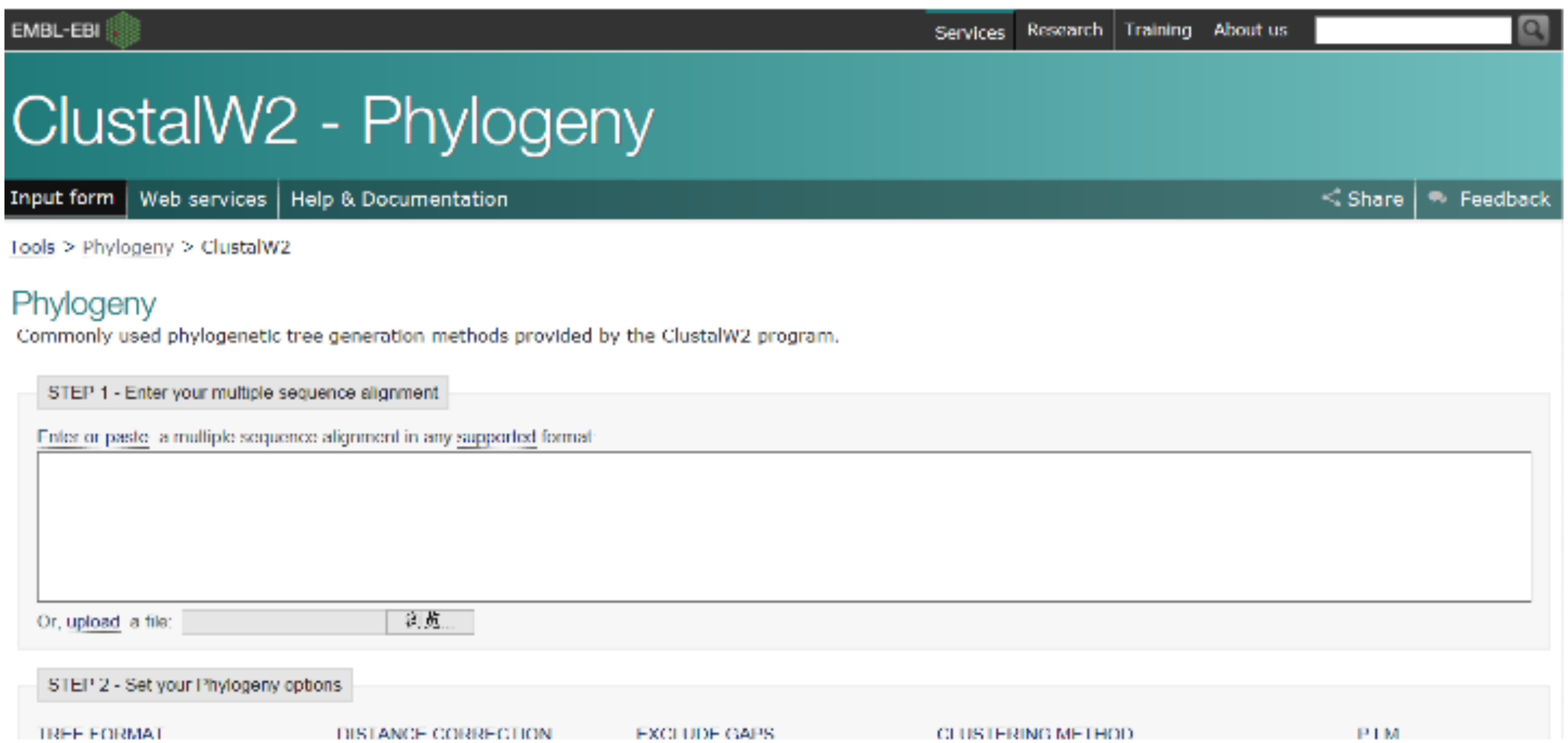


Clustal Omega (http://www.clustal.org/omega/)

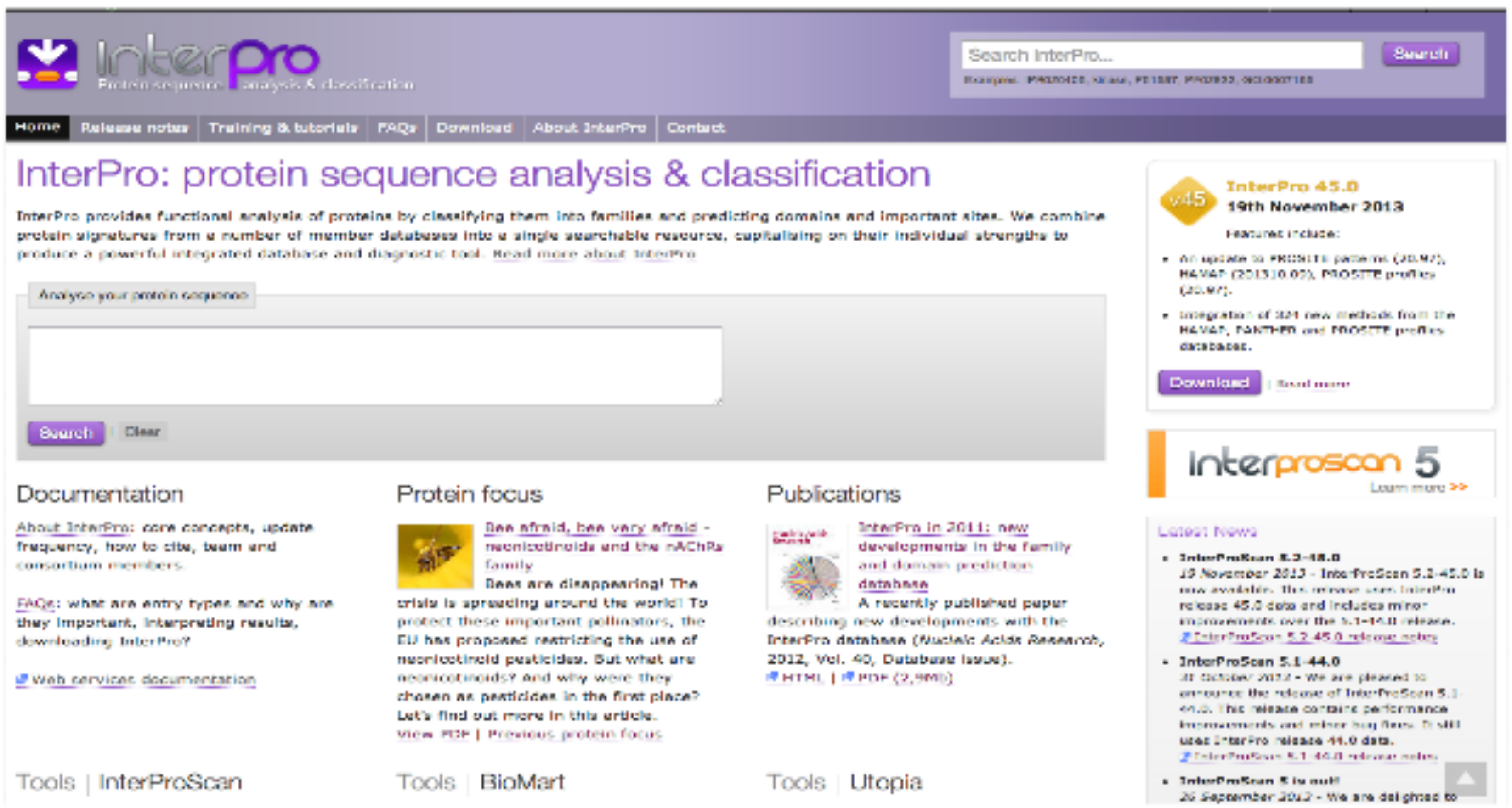


ClustalW2 – Phylogeny

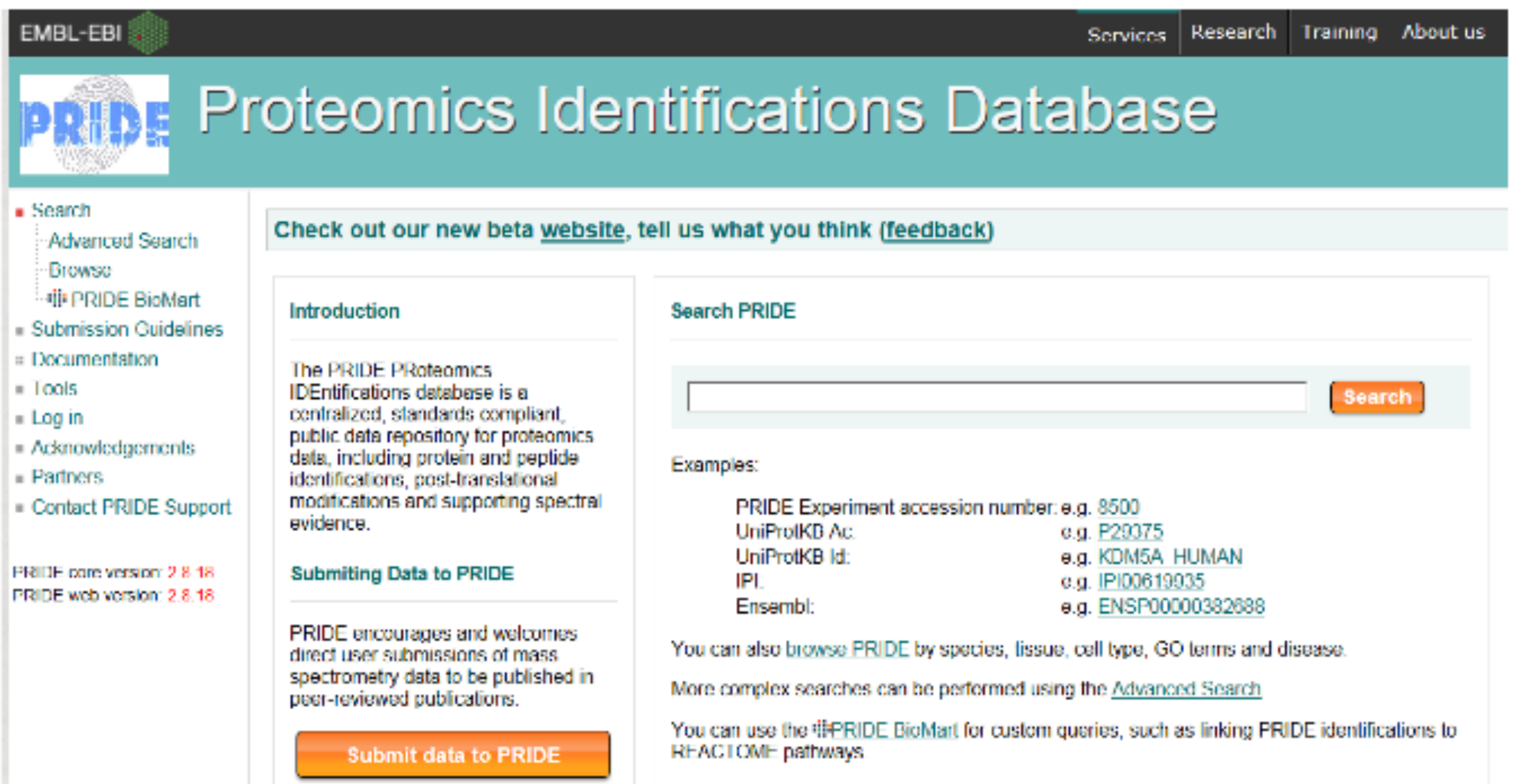
(http://www.ebi.ac.uk/Tools/phylogeny/clustalw2_phylogeny/)



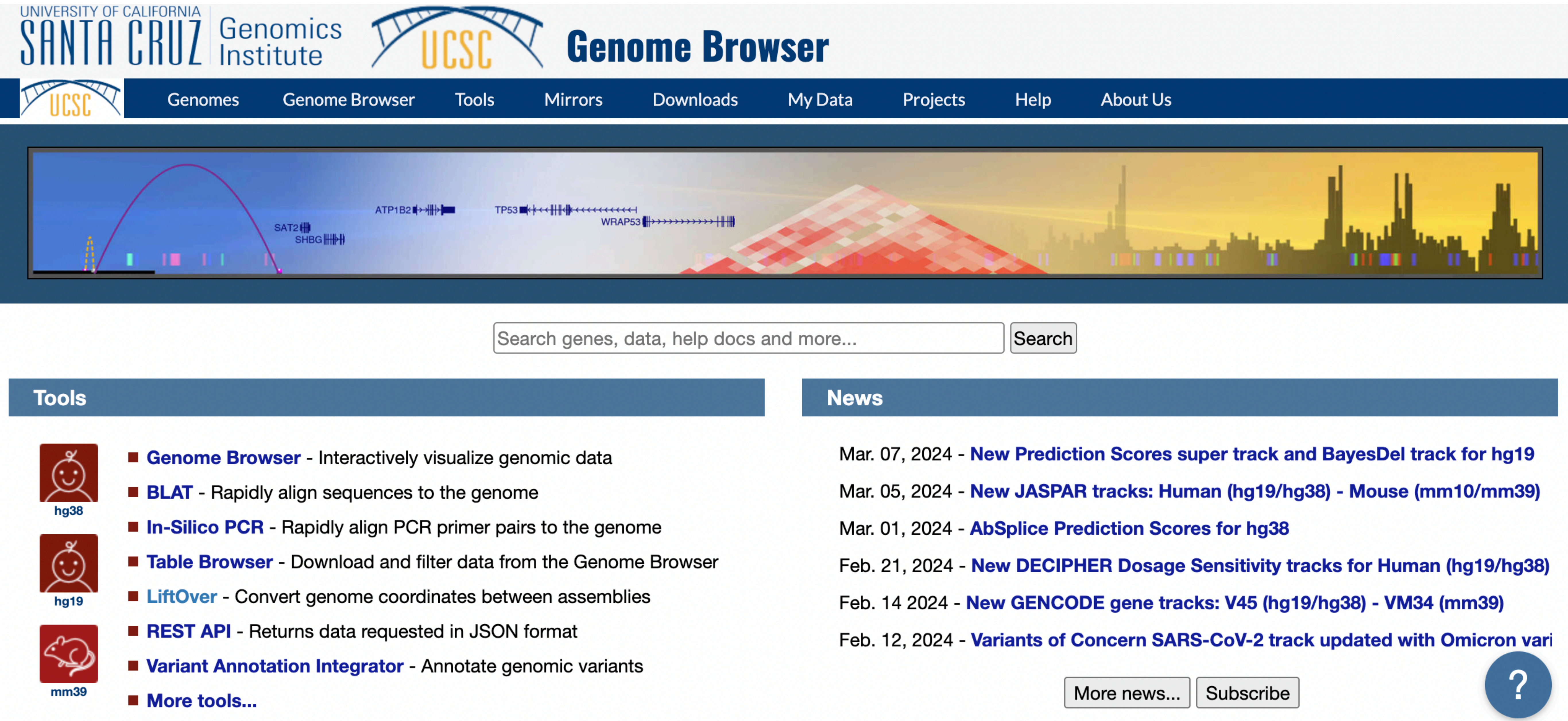
InterPro & InterProScan (http://www.ebi.ac.uk/interpro/)



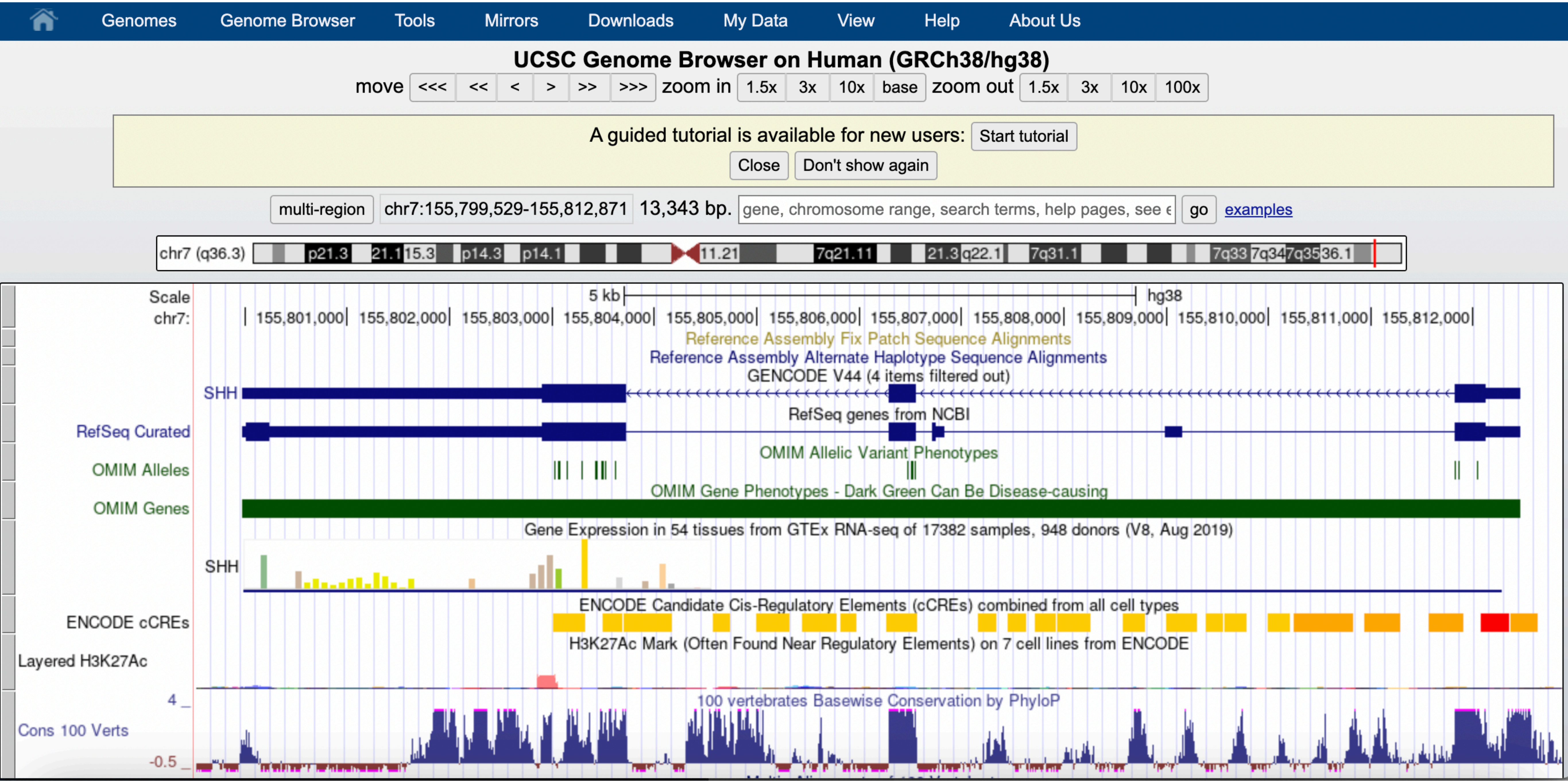
PRIDE (http://www.ebi.ac.uk/pride/)



UCSC Genome Bioinformatics (http://genome.ucsc.edu/)



UCSC Genome Browser (http://genome.ucsc.edu/)



The ENCODE Project: ENCyclopedia Of DNA Elements

Share Print

ENCODE

- **Overview**
- **Publications, Features and Press Releases**
- **Consortium Membership**
- **Data Release Policy**
- **Accessing ENCODE Data**
- **ENCODE Tutorials**
- **Common Cell Types**
- **Requests for Application (RFAs)**
- **Program Staff**

Follow the ENCODE Project on:

- Facebook
- Twitter

See Also:

Identification and analysis of functional elements in 1% of the human genome by the ENCODE pilot project PDF
Nature, June 13, 2007

Major Findings from The ENCODE Pilot Project
June 2007

The modENCODE Project

Grants Home

On Other Sites:

The ENCODE (ENCyclopedia Of DNA Elements) Project.

ENCODE data portal at UCSC

(http://genome.ucsc.edu/encode/)

ENCODE

Encyclopedia of DNA Elements

General

Resources & FAQ

Publications

Software Tools

Data Standards

Human

Downloads

Experiment Matrix

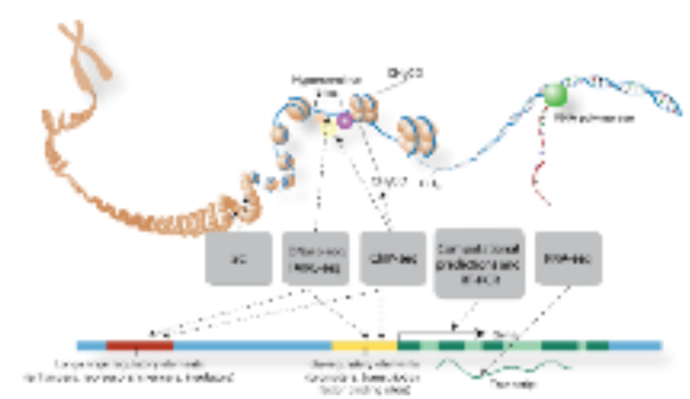
Search

Genome Browser (hg19)

Integrative Analysis

About ENCODE Data

The Encyclopedia of DNA Elements (ENCODE) Consortium is an international collaboration of research groups funded by the National Human Genome Research Institute (NHGRI). The goal of ENCODE is to build a comprehensive parts list of functional elements in the human genome, including elements that act at the protein and RNA levels, and regulatory elements that control cells and circumstances in which a gene is active.



Click to enlarge

ENCODE data are now available for the entire human genome. ***All ENCODE data are free and available for immediate use via*** :

- Search for displayable tracks and downloadable files
- Download of data files
- Visualization in the UCSC Genome Browser (ENCODE data marked with the NHGRI logo)
- Data mining with the UCSC Table Browser and other UCSC Genome Bioinformatics tools

To search for ENCODE data related to your area of interest and set up a browser view, use the UCSC Experiment Matrix or Track Search tool (*Advanced* features). The Experiment List (Human) and Experiment List (Mouse) links provide comprehensive listings of ENCODE data that is released or in preparation.

UCSC *In-Silico* PCR (http://genome.ucsc.edu/cgi-bin/hgPcr)

Genomes | Genome Browser | Tools | Mirrors | Downloads | My Data | About Us | Help

UCSC In-Silico PCR

Genome:	Assembly:	Target:	Forward Primer:	Reverse Primer:	
Human	Feb. 2009 (GRCh37/hg19)	genome assembly			submit

Max Product Size: 4000   Min Perfect Match: 15   Min Good Match: 15   Flip Reverse Primer: ☐

About In-Silico PCR

In-Silico PCR searches a sequence database with a pair of PCR primers, using an indexing strategy for fast performance.

Configuration Options

Genome and Assembly - The sequence database to search.
Target - If available, choose to query transcribed sequences.
Forward Primer - Must be at least 15 bases in length.
Reverse Primer - On the opposite strand from the forward primer. Minimum length of 15 bases.
Max Product Size - Maximum size of amplified region.
Min Perfect Match - Number of bases that match exactly on 3' end of primers. Minimum match size is 15.
Min Good Match - Number of bases on 3' end of primers where at least 2 out of 3 bases match.
Flip Reverse Primer - Invert the sequence order of the reverse primer and complement it.

Output

When successful, the search returns a sequence output file in fasta format containing all sequence in the database that lie between and include the primer pair. The fasta header describes the region in the database and the primers. The fasta body is capitalized in areas where the primer sequence matches the database sequence and in lower-case elsewhere. Here is an example from human:

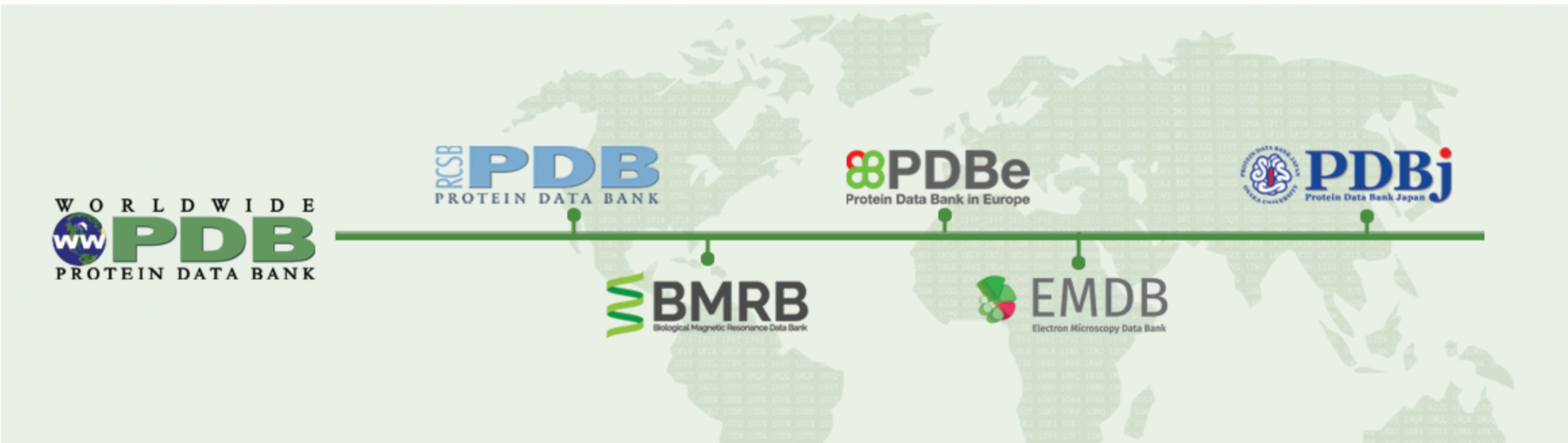
```
>chr22:31000551+31001000  TAACAGATTGATGATGCATGAAATGGG CCCATGAGTGGCTCCTAAAGCAGCTGC
TtACAGATTGATGATGCATGAAATGGGgggtggccaggggtgggggtga
```

PDB (http://www.rcsb.org/)

PDB

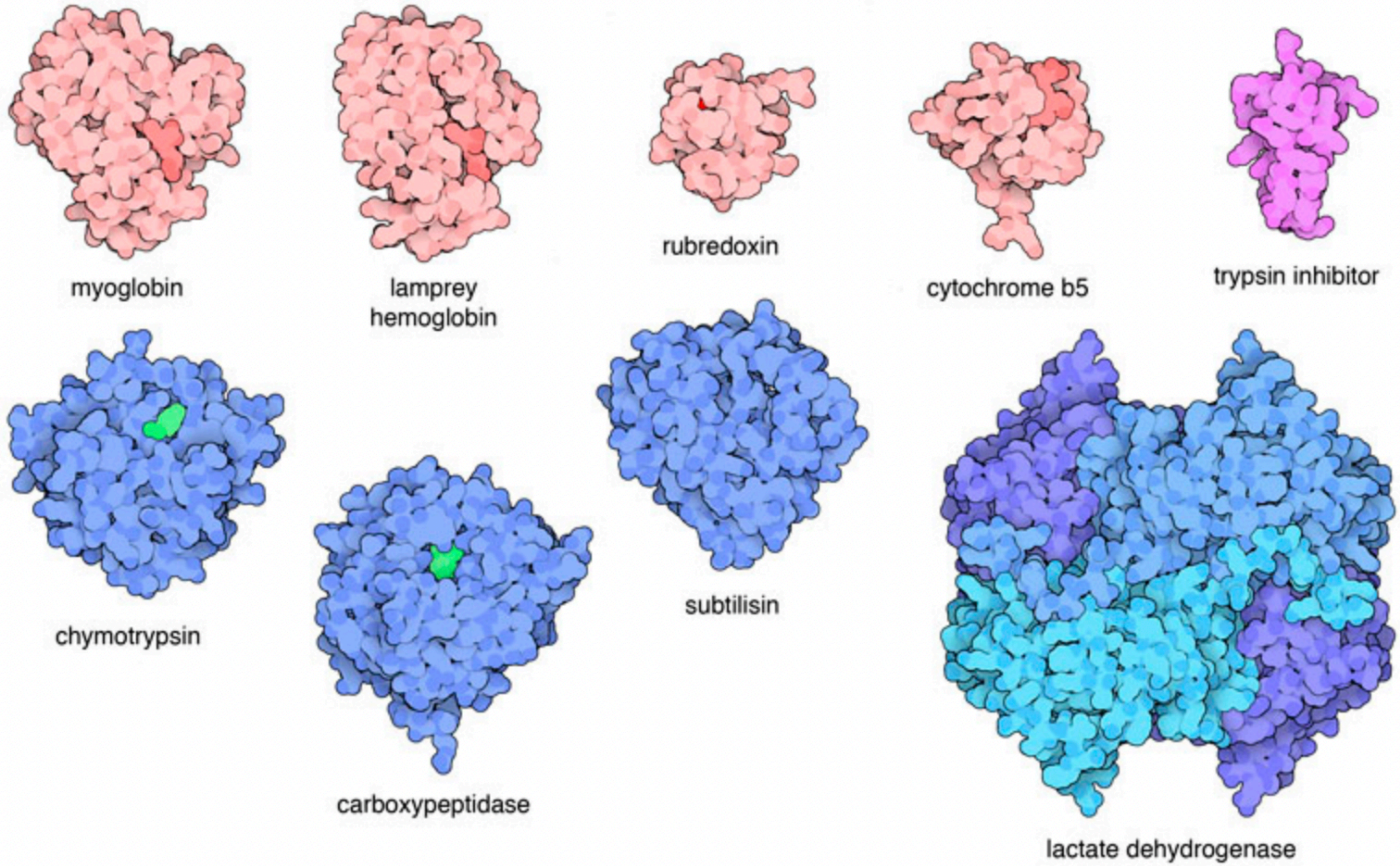
Protein Data Bank (PDB) archive of 3D structure data for biological molecules (proteins, DNA, RNA).

Currently includes > 1TB of structure data, archived world-wide.



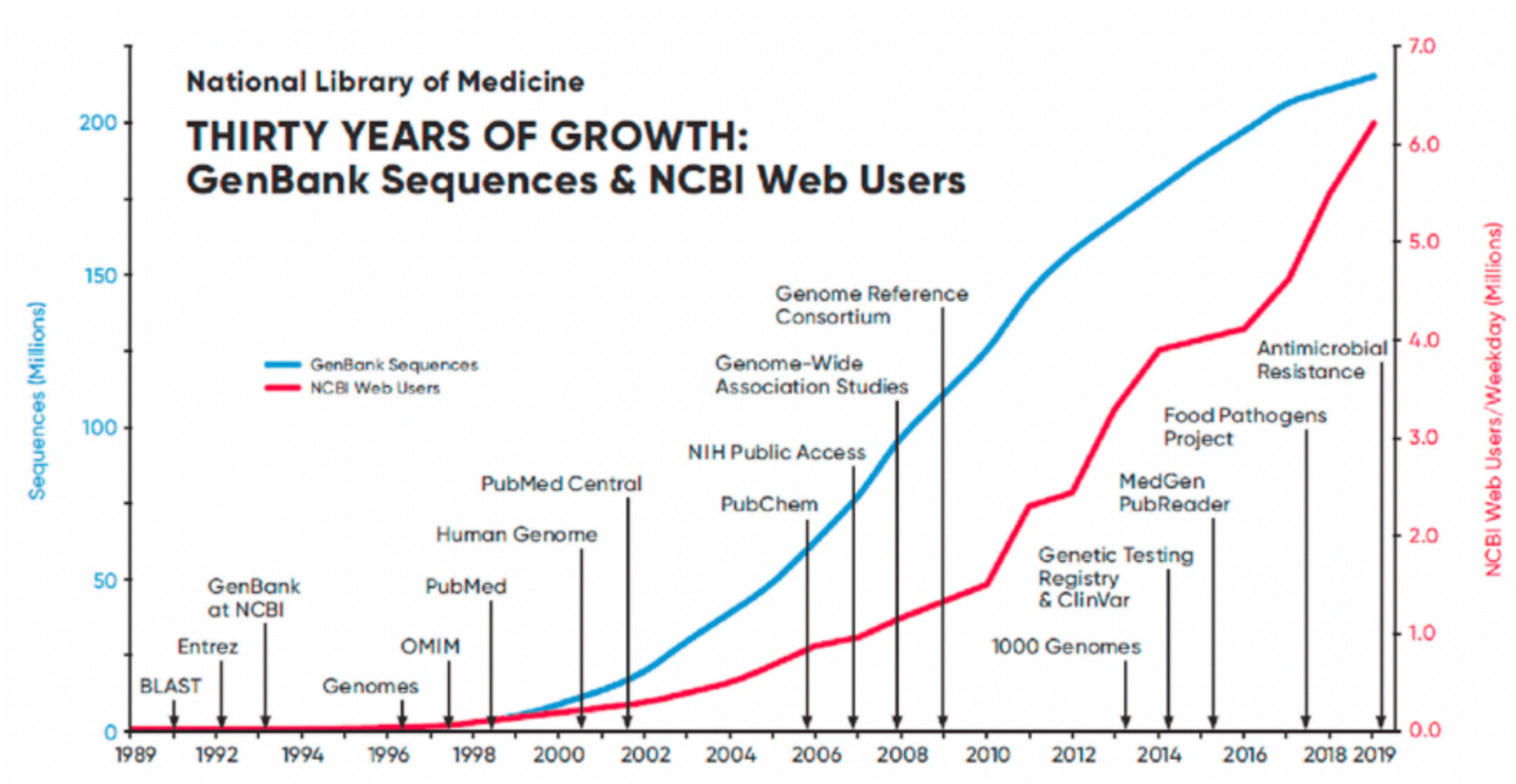
The Protein Data Bank, or PDB, is a vital repository for 3D structure data of biological molecules. We'll delve into the significance of PDB, its role in advancing structural biology, and the substantial volume of data it currently archives.

Protein Data Bank in 1973



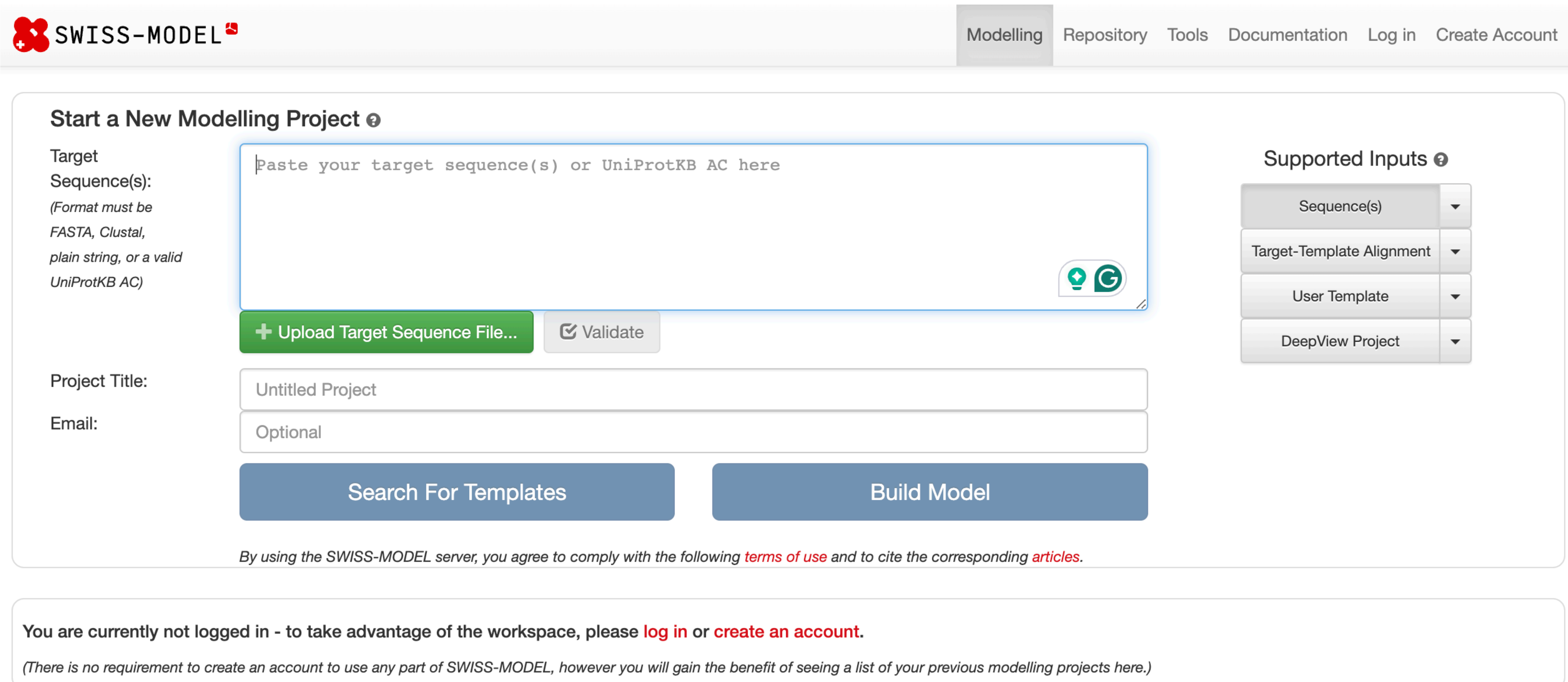
GenBank

An annotated collection of all publicly available DNA sequences, which comprises the DNA DataBank of Japan (DDBJ), the European Nucleotide Archive (ENA), and GenBank at NCBI



GenBank stands as a critical resource for DNA sequences. It collaborates with other databases, such as DDBJ and ENA, to provide a comprehensive collection of publicly available DNA sequences.

https://swissmodel.expasy.org/interactive



SWISS-MODEL Repository

(http://swissmodel.expasy.org/repository/?pid=smr01&zid=async)

The SWISS-MODEL Repository is a database of annotated 3D protein structure models generated by the SWISS-MODEL homology-modelling pipeline.

Bienert S, Waterhouse A, de Beer TA, Tauriello G, Studer G, Bordoli L, Schwede T (2017). The SWISS-MODEL Repository - new features and functionality *Nucleic Acids Res.* 45(D1):D313-D319. doi>

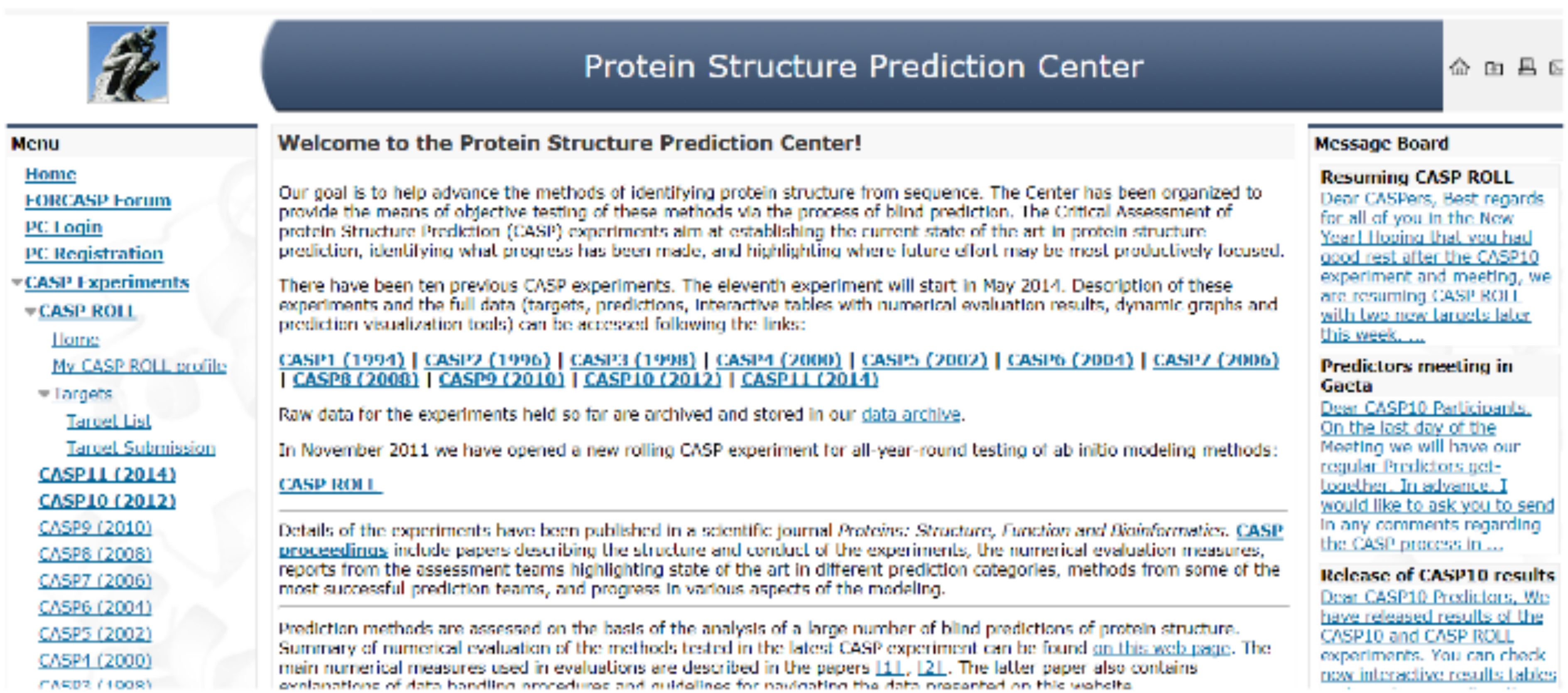
The aim of the SWISS-MODEL Repository is to provide access to an up-to-date collection of annotated 3D protein models generated by automated homology modelling for relevant model organisms and experimental structure information for all sequences in UniProtKB. Regular updates ensure that target coverage is complete, that models are built using the most recent sequence and template structure databases, and that improvements in the underlying modelling pipeline are fully utilised. It also allows users to assess the quality of the models using the latest QMEAN results. If a sequence has not been modelled, the user can build models interactively via the SWISS-MODEL workspace.

Currently the repository contains 2,543,873 models from SWISS-MODEL for UniProtKB targets as well as 203,493 structures from PDB with mapping to UniProtKB.

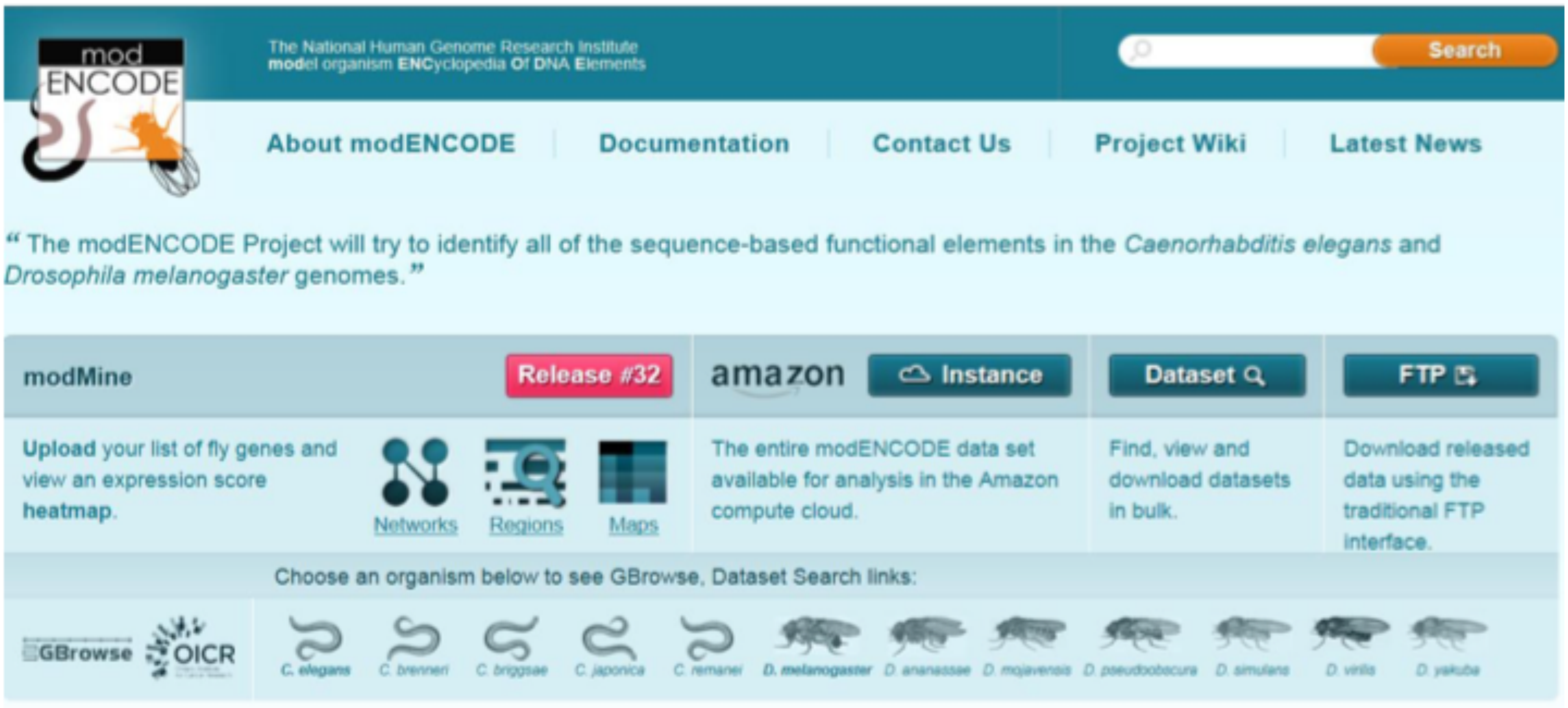
We currently provide models for the reference proteomes of the following model organisms, based on UniProtKB release 2024_01. If you want to download a large number of models, please contact us.

	Proteome Size	Sequences Modelled	Models	Seq Coverage	Metadata (Homology models *and* experimental structures)	Coordinates (Homology models only)
Homo sapiens	20,597	17,280	42,868		*14.1 MB*	*5.5 GB*
Mus musculus	21,701	18,785	43,406		*7.6 MB*	*3.9 GB*
Caenorhabditis elegans	19,827	12,744	22,978		*3.5 MB*	*1.6 GB*
Escherichia coli	4,403	3,675	6,169		*2.0 MB*	*482.5 MB*
Arabidopsis thaliana	27,448	20,179	37,235		*5.4 MB*	*2.5 GB*
Drosophila melanogaster	13,824	10,000	19,691		*3.0 MB*	*1.7 GB*
Saccharomyces cerevisiae	6,060	4,631	8,516		*2.2 MB*	*598.6 MB*

CASP (http://www.predictioncenter.org/)



modENCODE (http://www.modencode.org/)



Rfam (http://rfam.sanger.ac.uk)


wellcome trust sanger institute

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Rfam

Go

Please note: this site relies heavily on the use of javascript. Without a javascript-enabled browser, this site will not function correctly. Please enable javascript and reload the page, or switch to a different browser.

Rfam 11.0 (August 2012, 2208 families)

The Rfam database is a collection of RNA families, each represented by **multiple sequence alignments, consensus secondary structures** and **covariance models (CMs)**. More...

QUICK LINKS	YOU CAN FIND DATA IN RFAM IN VARIOUS WAYS...
SEQUENCE SEARCH	Analyze your RNA sequence for Rfam matches
VIEW AN RFAM FAMILY	View Rfam family annotation and alignments
VIEW AN RFAM CLAN	View Rfam clan details
KEYWORD SEARCH	Query Rfam by keywords
TAXONOMY SEARCH	Fetch families or sequences by NCBI taxonomy
JUMP TO	Go Example Enter any type of accession or ID to jump to the page for a Rfam family, sequence or genome

Or view the help pages for more information

Citing Rfam

If you find Rfam useful, please consider citing the references that describe this work:

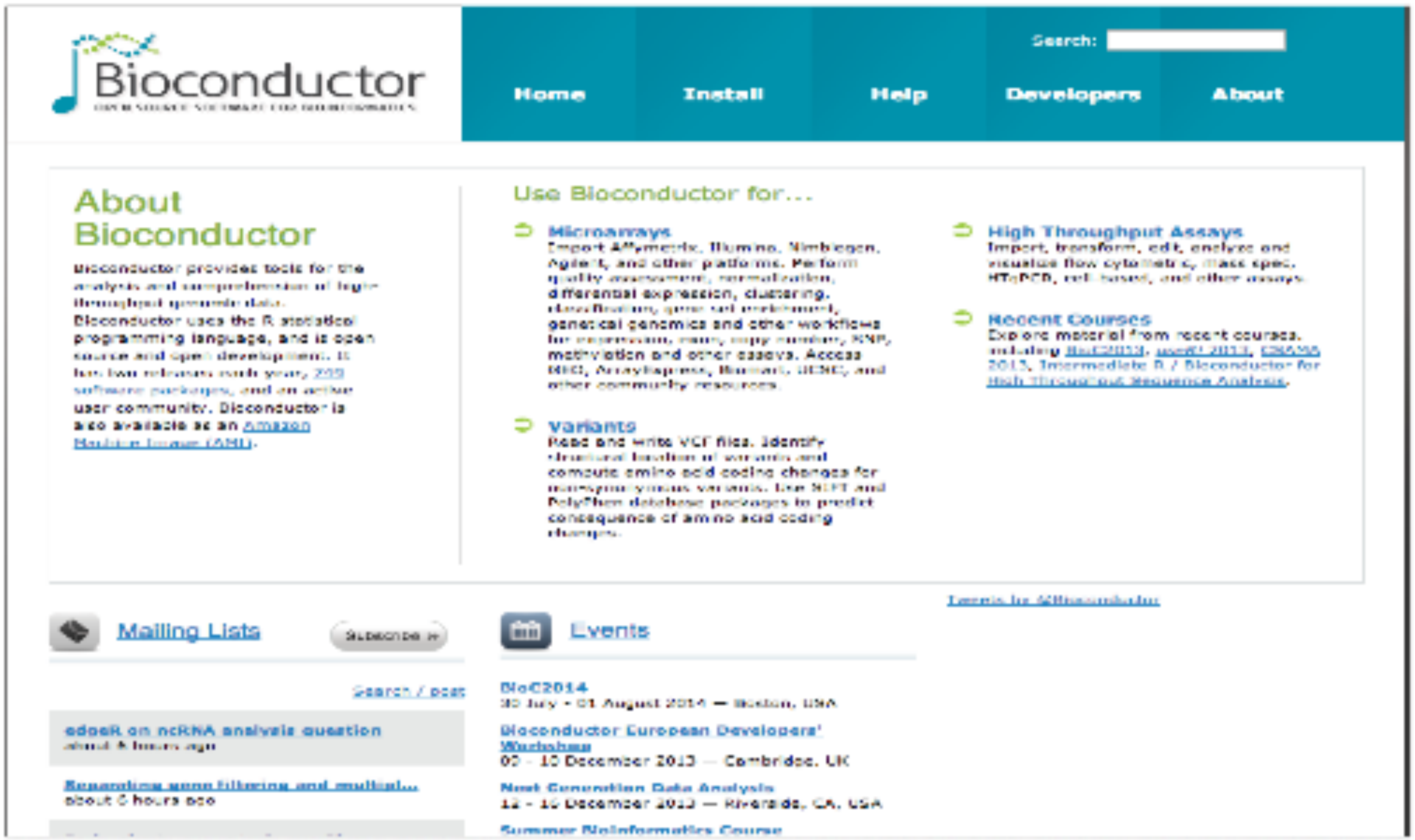
Rfam 11.0: 10 years of RNA families. S.W. Burge, J. Daub, R. Eberhardt, J. Tate, L. Barquist, E.P. Nawrocki, S.R. Eddy, P.P. Gardner, A. Bateman.
Nucleic Acids Research (2012) doi: 10.1093/nar/gks1005

Mirrors

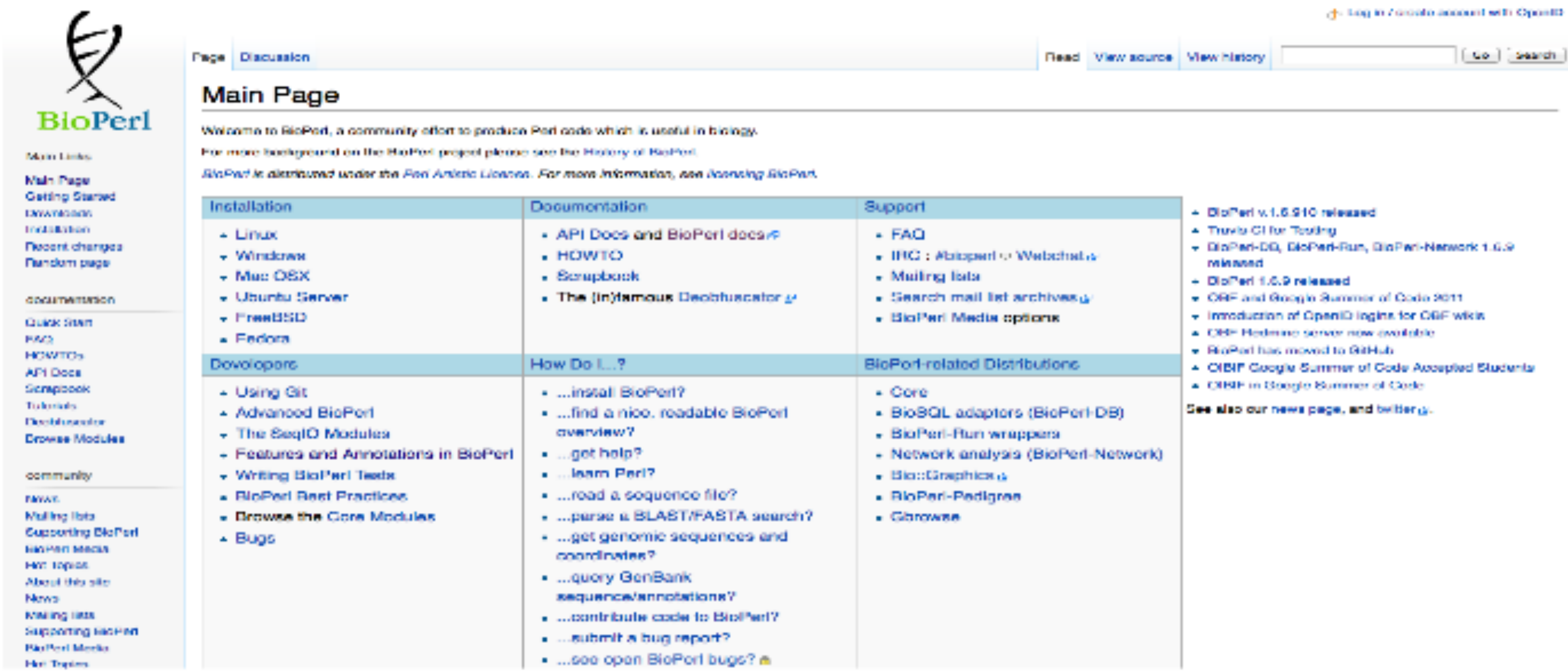
The following are official Rfam mirror sites:

- WTSI, UK
- JFRC, USA

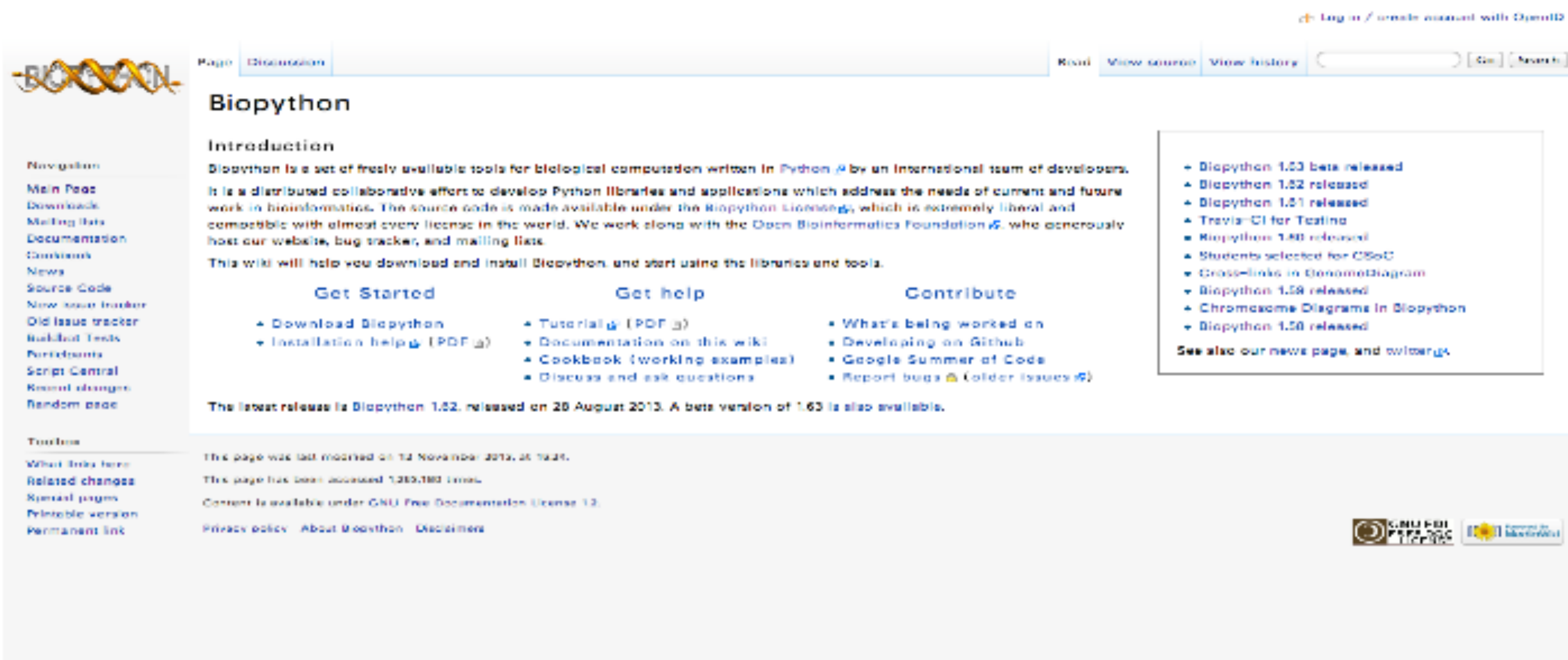
Bioconductor (http://bioconductor.org/)



BioPerl (http://bioperl.org/)



BioPython (http://biopython.org)



THANK YOU